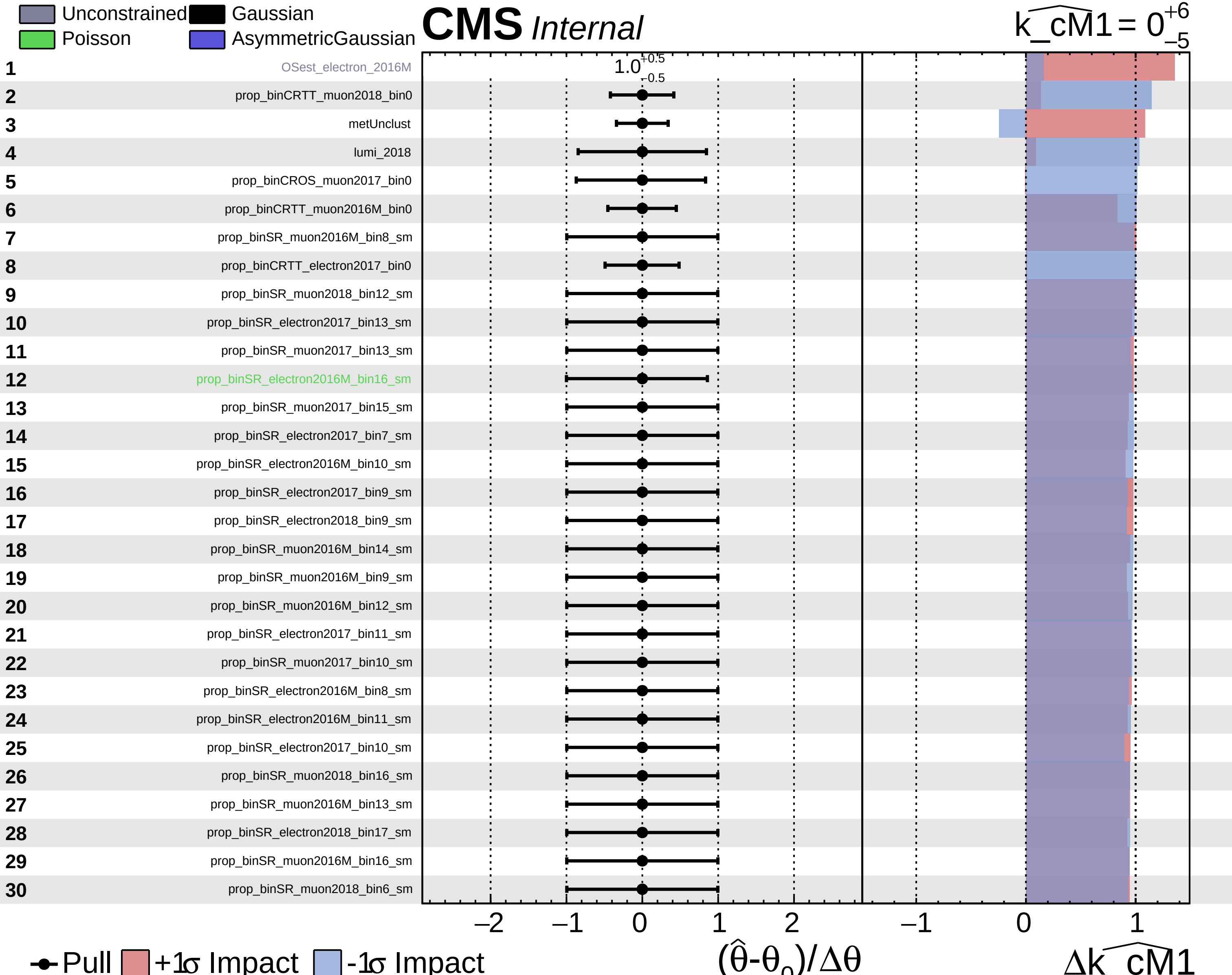
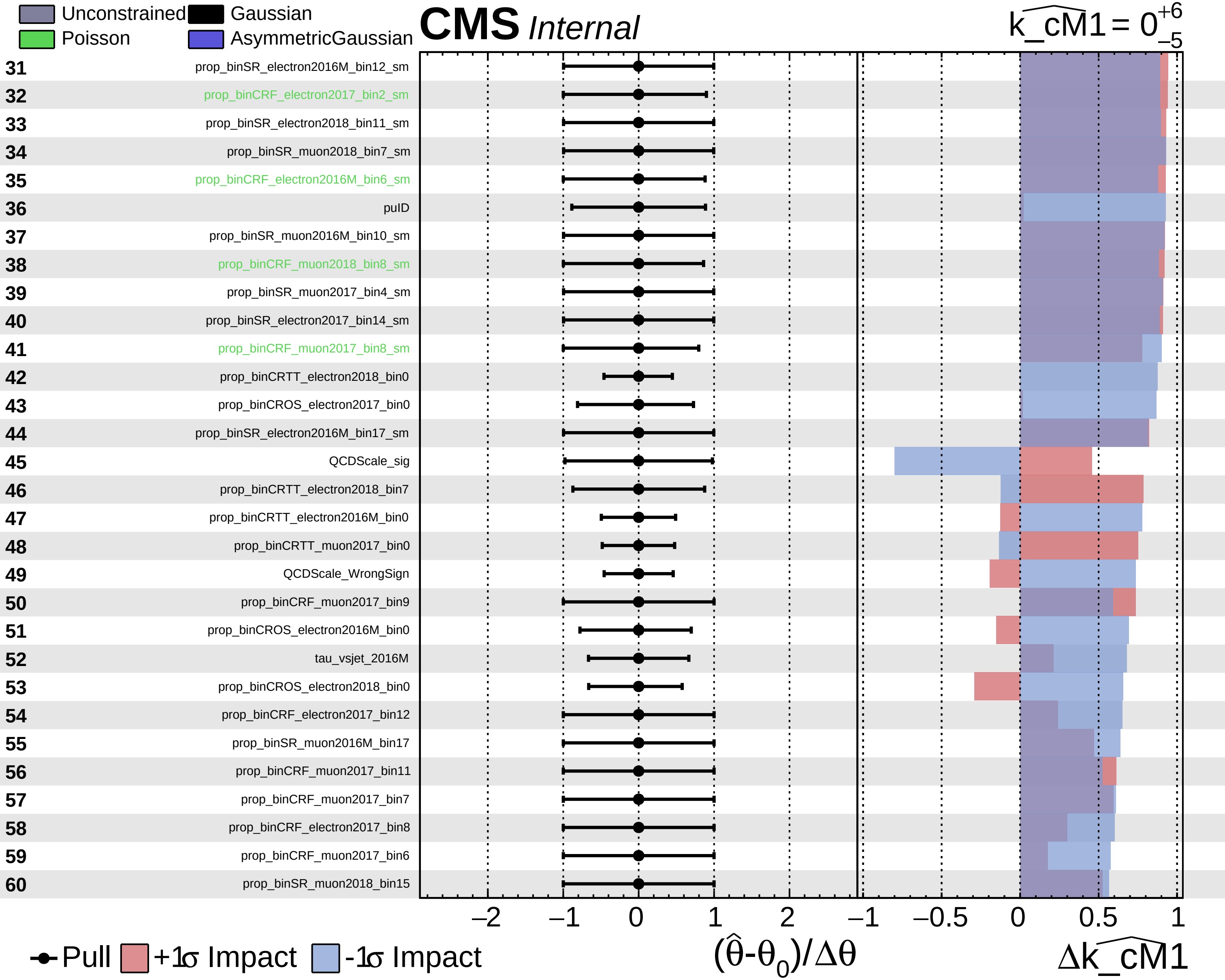
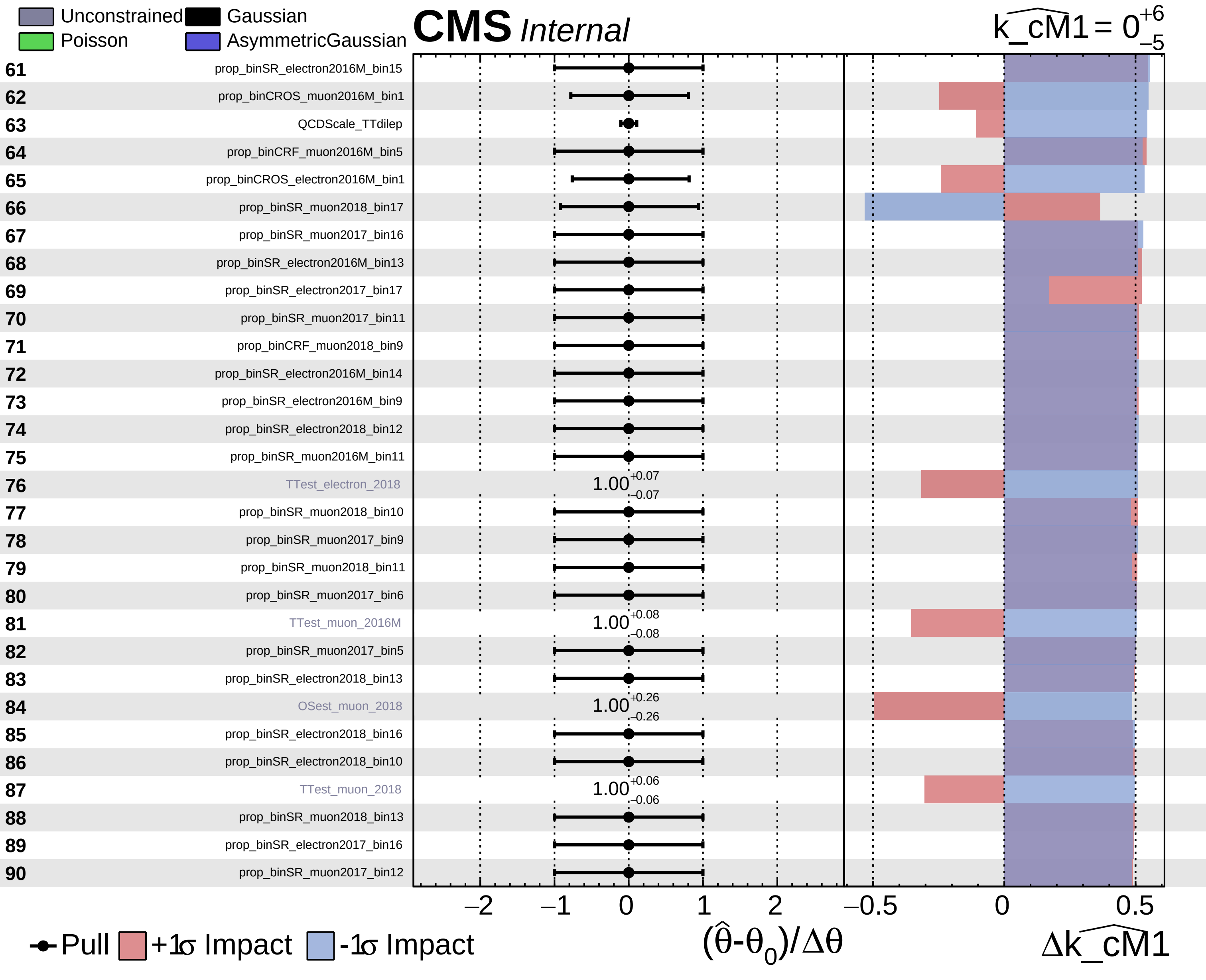
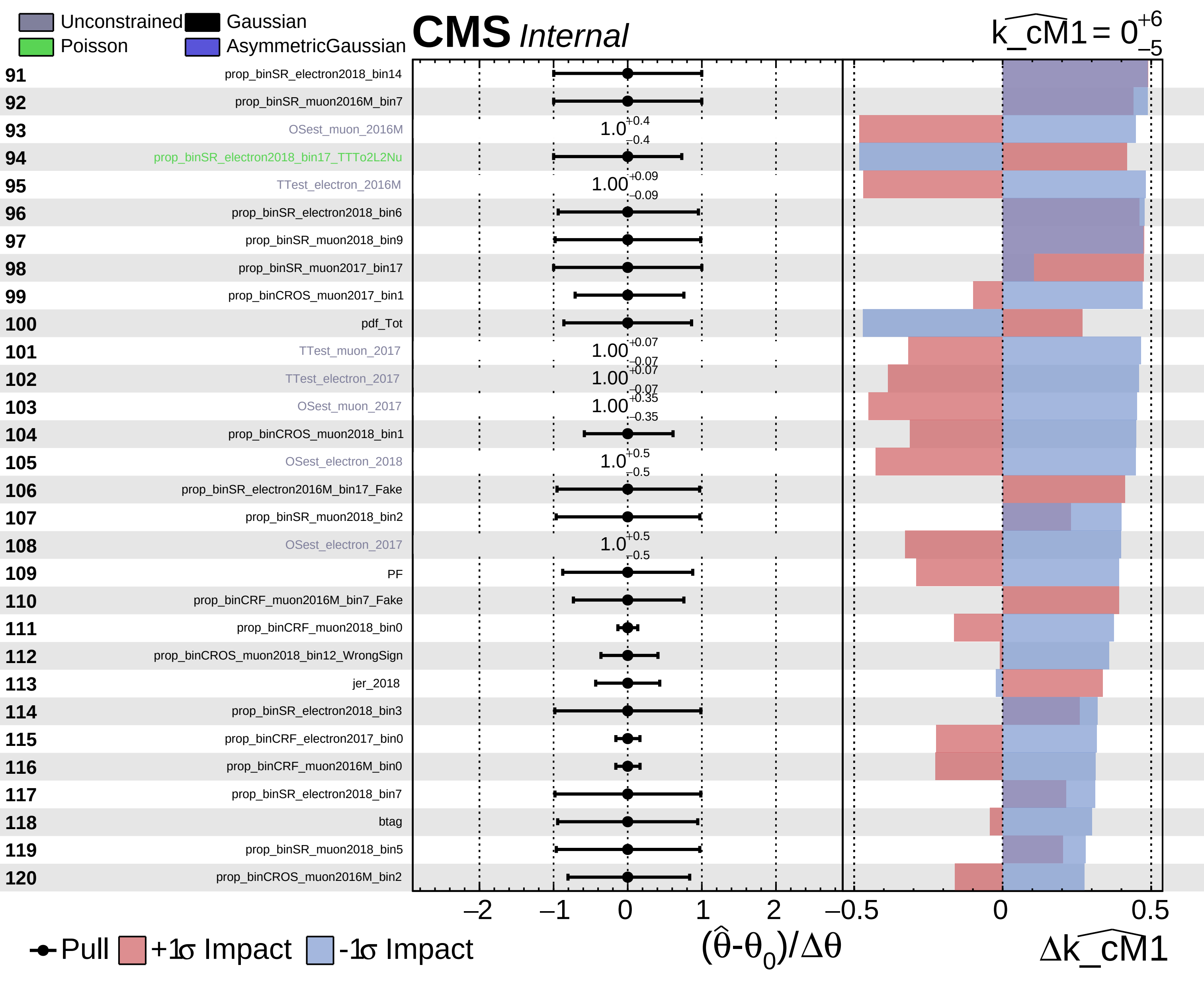


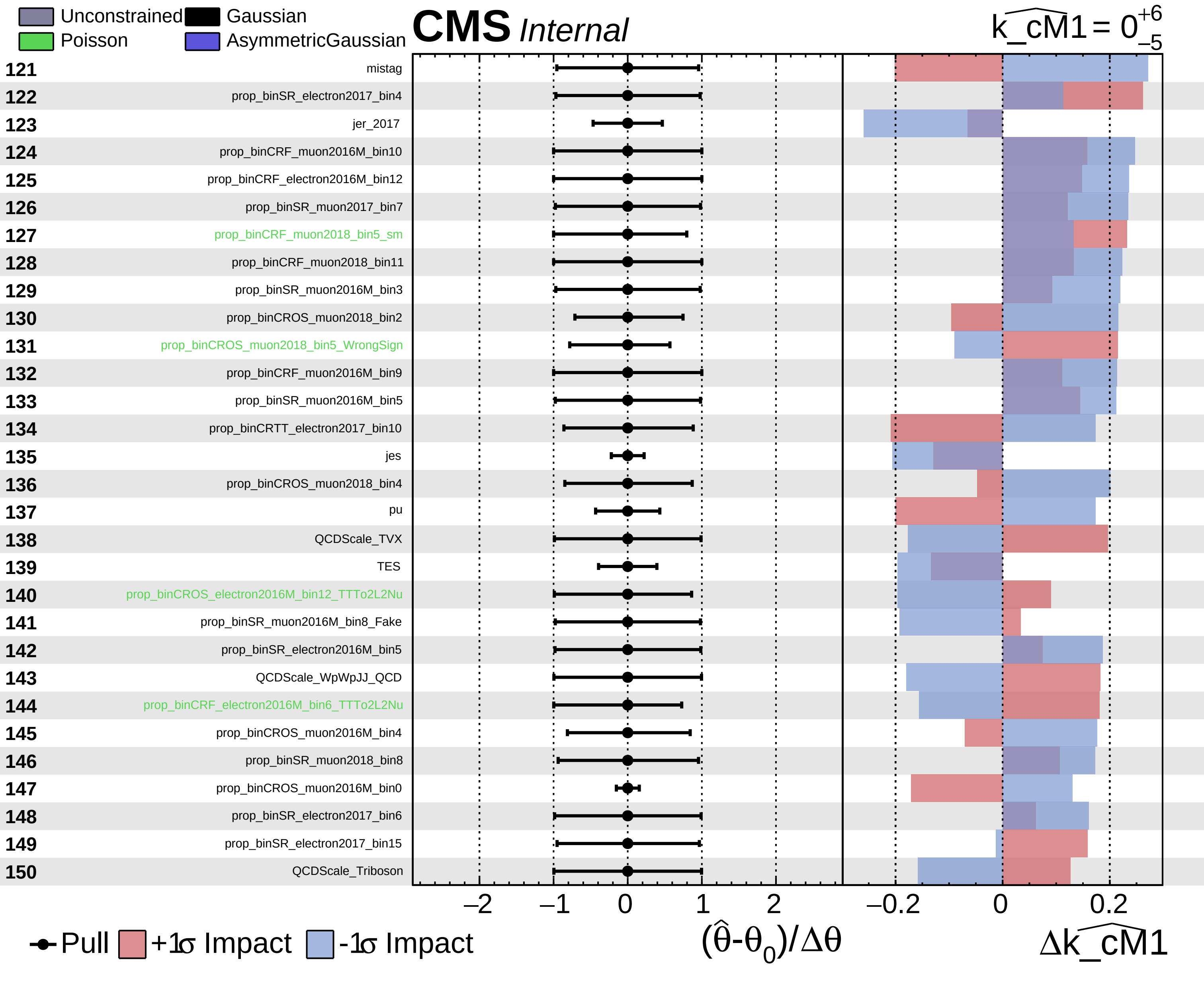
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0^{+6}_{-5}
1 OSest_electron_2016M
$1.0^{+0.5}_{-0.5}$
2 prop_binCRTT_muon2018_bin0
3 metUnclust
4 lumi_2018
5 prop_binCROS_muon2017_bin0
6 prop_binCRTT_muon2016M_bin0
7 prop_binSR_muon2016M_bin8_sm
8 prop_binCRTT_electron2017_bin0
9 prop_binSR_muon2018_bin12_sm
10 prop_binSR_electron2017_bin13_sm
11 prop_binSR_muon2017_bin13_sm
12 prop_binSR_electron2016M_bin16_sm
13 prop_binSR_muon2017_bin15_sm
14 prop_binSR_electron2017_bin7_sm
15 prop_binSR_electron2016M_bin10_sm
16 prop_binSR_electron2017_bin9_sm
17 prop_binSR_electron2018_bin9_sm
18 prop_binSR_muon2016M_bin14_sm
19 prop_binSR_muon2016M_bin9_sm
20 prop_binSR_muon2016M_bin12_sm
21 prop_binSR_electron2017_bin11_sm
22 prop_binSR_muon2017_bin10_sm
23 prop_binSR_electron2016M_bin8_sm
24 prop_binSR_electron2016M_bin11_sm
25 prop_binSR_electron2017_bin10_sm
26 prop_binSR_muon2018_bin16_sm
27 prop_binSR_muon2016M_bin13_sm
28 prop_binSR_electron2018_bin17_sm
29 prop_binSR_muon2016M_bin16_sm
30 prop_binSR_muon2018_bin6_sm
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1

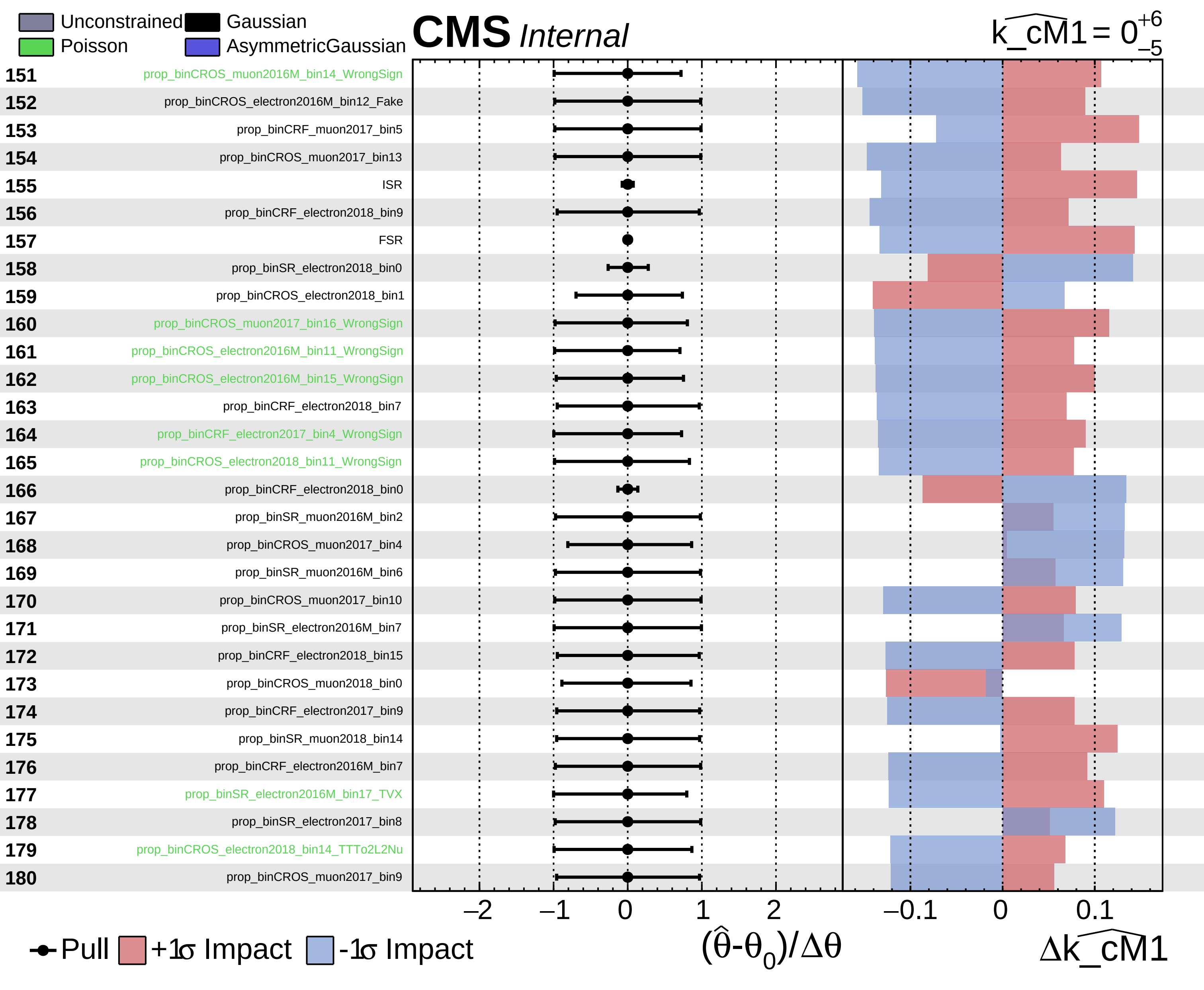

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
31 prop_binSR_electron2016M_bin12_sm
32 prop_binCRF_electron2017_bin2_sm
33 prop_binSR_electron2018_bin11_sm
34 prop_binSR_muon2018_bin7_sm
35 prop_binCRF_electron2016M_bin6_sm
36 puID
37 prop_binSR_muon2016M_bin10_sm
38 prop_binCRF_muon2018_bin8_sm
39 prop_binSR_muon2017_bin4_sm
40 prop_binSR_electron2017_bin14_sm
41 prop_binCRF_muon2017_bin8_sm
42 prop_binCRTT_electron2018_bin0
43 prop_binCROS_electron2017_bin0
44 prop_binSR_electron2016M_bin17_sm
45 QCDScale_sig
46 prop_binCRTT_electron2018_bin7
47 prop_binCRTT_electron2016M_bin0
48 prop_binCRTT_muon2017_bin0
49 QCDScale_WrongSign
50 prop_binCRF_muon2017_bin9
51 prop_binCROS_electron2016M_bin0
52 tau_vsjet_2016M
53 prop_binCROS_electron2018_bin0
54 prop_binCRF_electron2017_bin12
55 prop_binSR_muon2016M_bin17
56 prop_binCRF_muon2017_bin11
57 prop_binCRF_muon2017_bin7
58 prop_binCRF_electron2017_bin8
59 prop_binCRF_muon2017_bin6
60 prop_binSR_muon2018_bin15
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

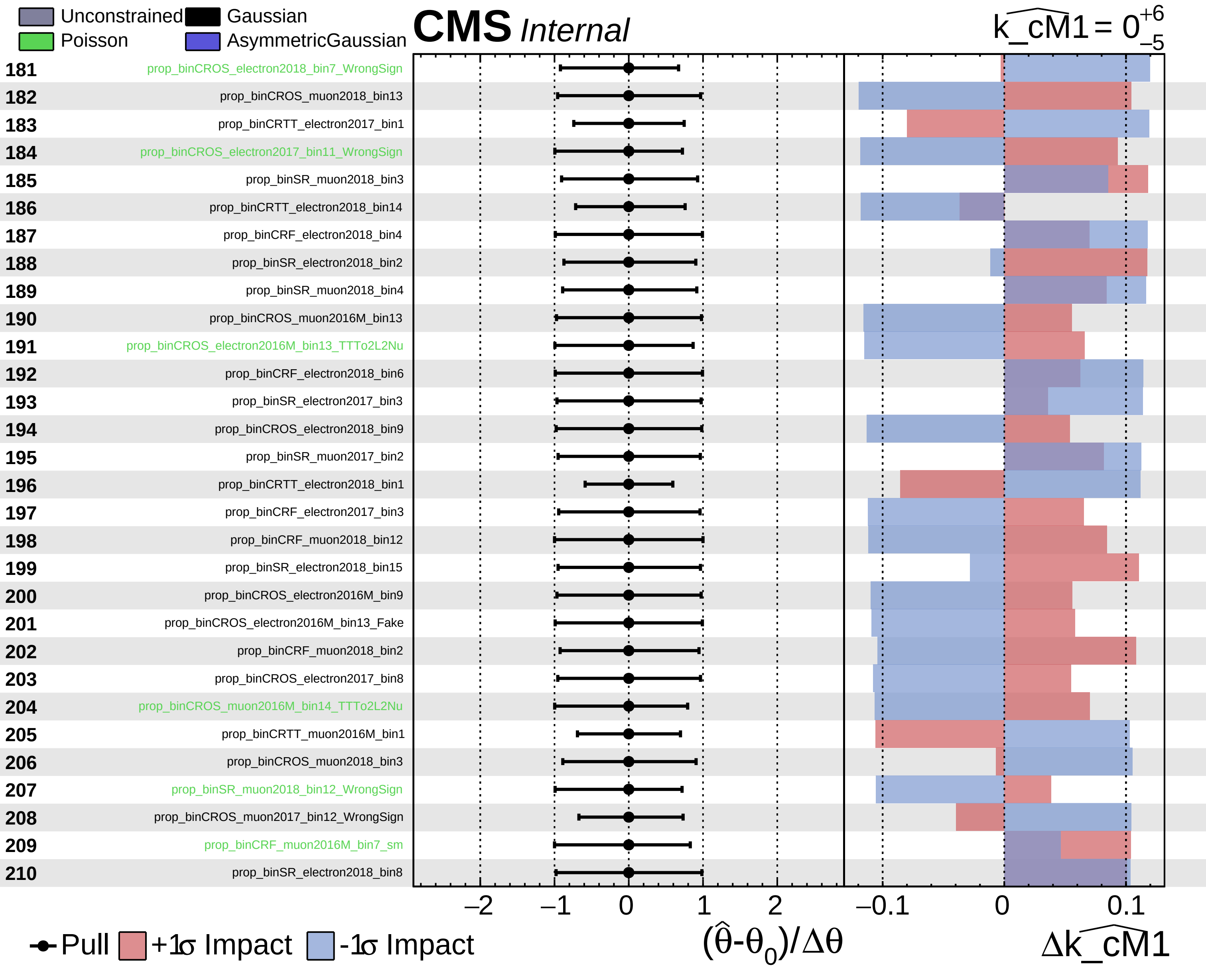


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
61 prop_binSR_electron2016M_bin15
62 prop_binCROS_muon2016M_bin1
63 QCDScale_TTdilep
64 prop_binCRF_muon2016M_bin5
65 prop_binCROS_electron2016M_bin1
66 prop_binSR_muon2018_bin17
67 prop_binSR_muon2017_bin16
68 prop_binSR_electron2016M_bin13
69 prop_binSR_electron2017_bin17
70 prop_binSR_muon2017_bin11
71 prop_binCRF_muon2018_bin9
72 prop_binSR_electron2016M_bin14
73 prop_binSR_electron2016M_bin9
74 prop_binSR_electron2018_bin12
75 prop_binSR_muon2016M_bin11
76 TTest_electron_2018 1.00 +0.07 −0.07
77 prop_binSR_muon2018_bin10
78 prop_binSR_muon2017_bin9
79 prop_binSR_muon2018_bin11
80 prop_binSR_muon2017_bin6
81 TTest_muon_2016M 1.00 +0.08 −0.08
82 prop_binSR_muon2017_bin5
83 prop_binSR_electron2018_bin13
84 OSest_muon_2018 1.00 +0.26 −0.26
85 prop_binSR_electron2018_bin16
86 prop_binSR_electron2018_bin10
87 TTest_muon_2018 1.00 +0.06 −0.06
88 prop_binSR_muon2018_bin13
89 prop_binSR_electron2017_bin16
90 prop_binSR_muon2017_bin12
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1

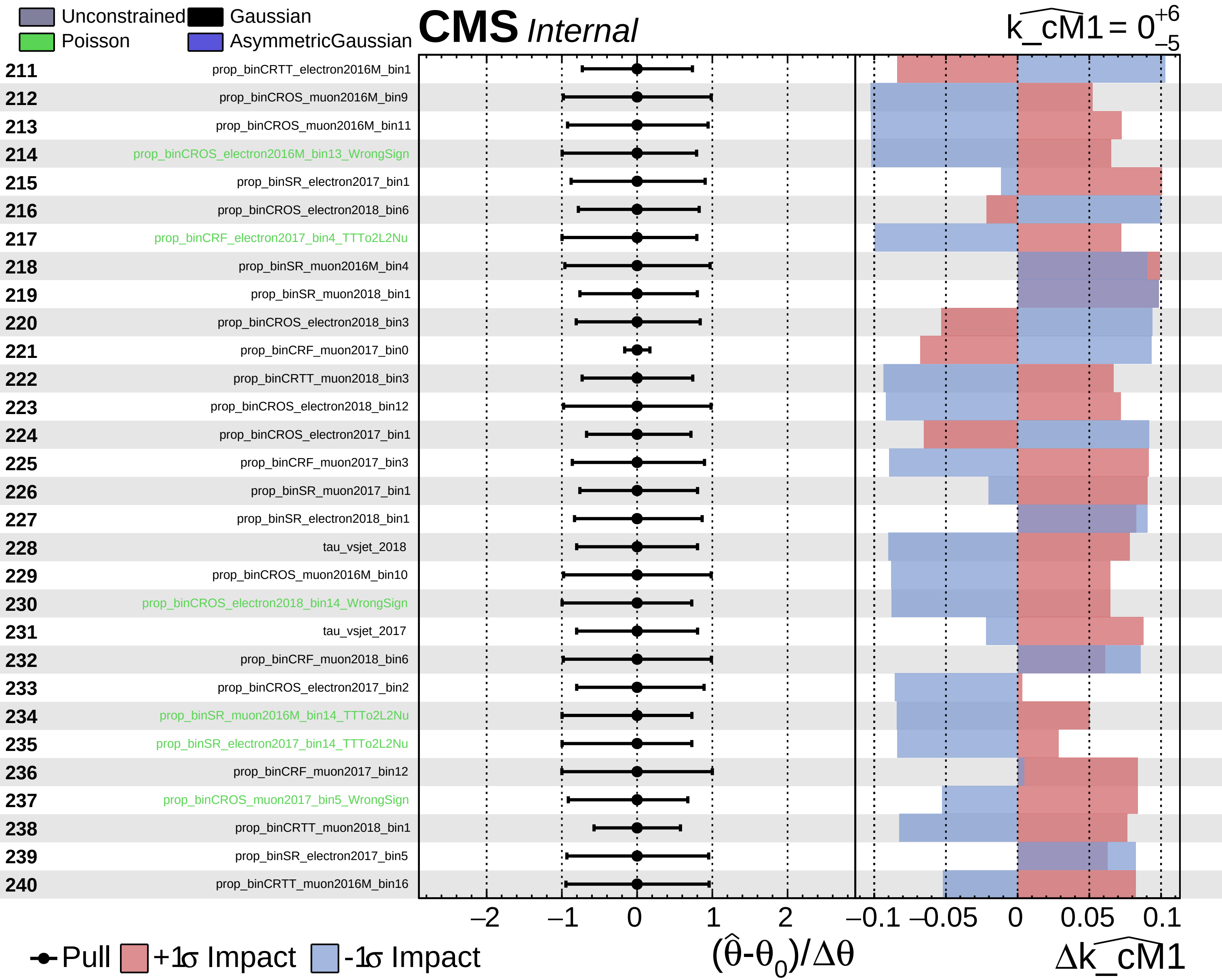

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
91 prop_binSR_electron2018_bin14
92 prop_binSR_muon2016M_bin7
93 OSest_muon_2016M 1.0 +0.4 −0.4
94 prop_binSR_electron2018_bin17_TTTo2L2Nu
95 TTest_electron_2016M 1.00 +0.09 −0.09
96 prop_binSR_electron2018_bin6
97 prop_binSR_muon2018_bin9
98 prop_binSR_muon2017_bin17
99 prop_binCROS_muon2017_bin1
100 pdf_Tot
101 TTest_muon_2017 1.00 +0.07 −0.07
102 TTest_electron_2017 1.00 +0.07 −0.07
103 OSest_muon_2017 1.00 +0.35 −0.35
104 prop_binCROS_muon2018_bin1
105 OSest_electron_2018 1.0 +0.5 −0.5
106 prop_binSR_electron2016M_bin17_Fake
107 prop_binSR_muon2018_bin2
108 OSest_electron_2017 1.0 +0.5 −0.5
109 PF
110 prop_binCRF_muon2016M_bin7_Fake
111 prop_binCRF_muon2018_bin0
112 prop_binCROS_muon2018_bin12_WrongSign
113 jer_2018
114 prop_binSR_electron2018_bin3
115 prop_binCRF_electron2017_bin0
116 prop_binCRF_muon2016M_bin0
117 prop_binSR_electron2018_bin7
118 btag
119 prop_binSR_muon2018_bin5
120 prop_binCROS_muon2016M_bin2
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1

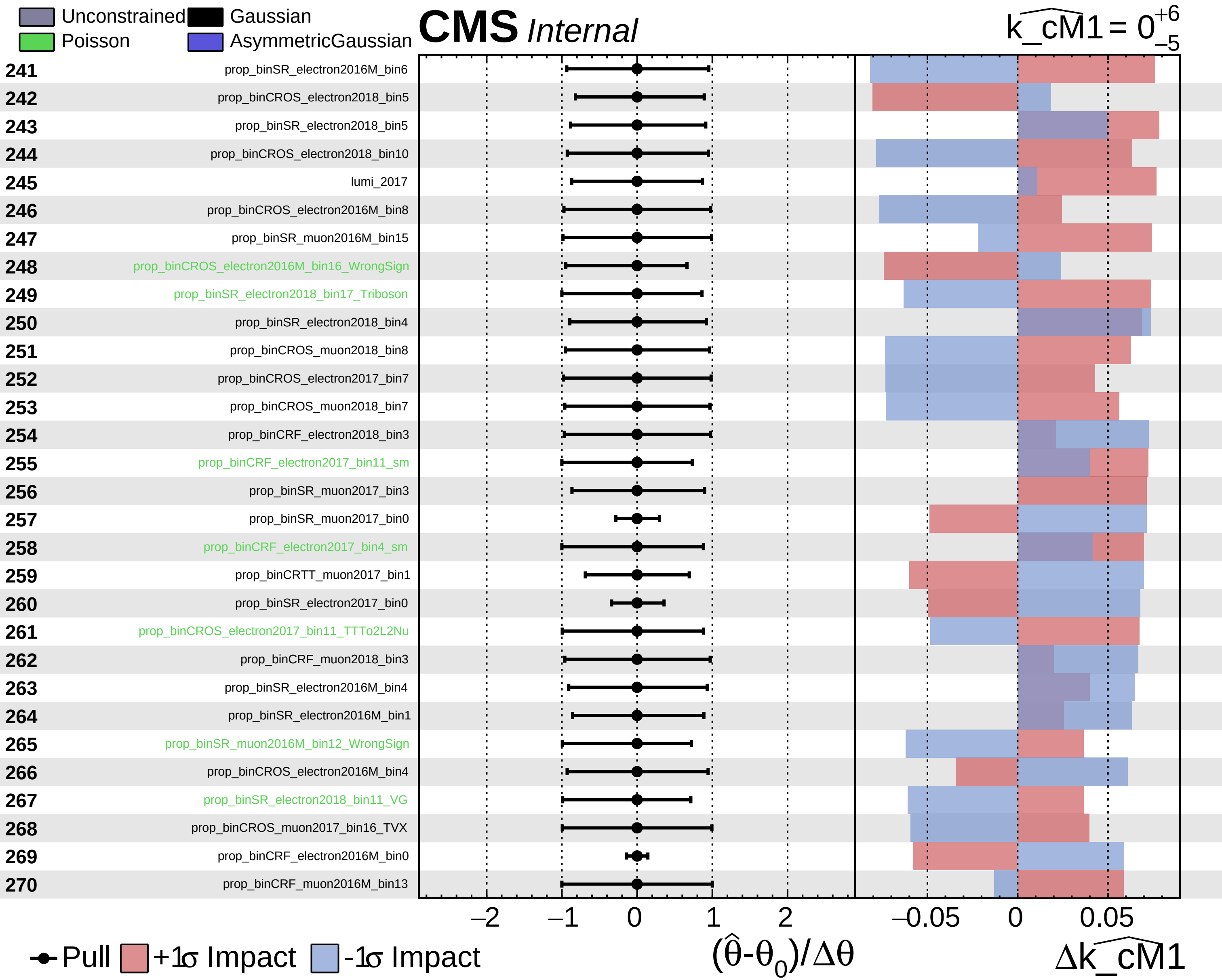

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
121 mistag
122 prop_binSR_electron2017_bin4
123 jer_2017
124 prop_binCRF_muon2016M_bin10
125 prop_binCRF_electron2016M_bin12
126 prop_binSR_muon2017_bin7
127 prop_binCRF_muon2018_bin5_sm
128 prop_binCRF_muon2018_bin11
129 prop_binSR_muon2016M_bin3
130 prop_binCROS_muon2018_bin2
131 prop_binCROS_muon2018_bin5_WrongSign
132 prop_binCRF_muon2016M_bin9
133 prop_binSR_muon2016M_bin5
134 prop_binCRTT_electron2017_bin10
135 jes
136 prop_binCROS_muon2018_bin4
137 pu
138 QCDScale_TVX
139 TES
140 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
141 prop_binSR_muon2016M_bin8_Fake
142 prop_binSR_electron2016M_bin5
143 QCDScale_WpWpJJ_QCD
144 prop_binCRF_electron2016M_bin6_TTTo2L2Nu
145 prop_binCROS_muon2016M_bin4
146 prop_binSR_muon2018_bin8
147 prop_binCROS_muon2016M_bin0
148 prop_binSR_electron2017_bin6
149 prop_binSR_electron2017_bin15
150 QCDScale_Triboson
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
151 prop_binCROS_muon2016M_bin14_WrongSign
152 prop_binCROS_electron2016M_bin12_Fake
153 prop_binCRF_muon2017_bin5
154 prop_binCROS_muon2017_bin13
155 ISR
156 prop_binCRF_electron2018_bin9
157 FSR
158 prop_binSR_electron2018_bin0
159 prop_binCROS_electron2018_bin1
160 prop_binCROS_muon2017_bin16_WrongSign
161 prop_binCROS_electron2016M_bin11_WrongSign
162 prop_binCROS_electron2016M_bin15_WrongSign
163 prop_binCRF_electron2018_bin7
164 prop_binCRF_electron2017_bin4_WrongSign
165 prop_binCROS_electron2018_bin11_WrongSign
166 prop_binCRF_electron2018_bin0
167 prop_binSR_muon2016M_bin2
168 prop_binCROS_muon2017_bin4
169 prop_binSR_muon2016M_bin6
170 prop_binCROS_muon2017_bin10
171 prop_binSR_electron2016M_bin7
172 prop_binCRF_electron2018_bin15
173 prop_binCROS_muon2018_bin0
174 prop_binCRF_electron2017_bin9
175 prop_binSR_muon2018_bin14
176 prop_binCRF_electron2016M_bin7
177 prop_binSR_electron2016M_bin17_TVX
178 prop_binSR_electron2017_bin8
179 prop_binCROS_electron2018_bin14_TTTo2L2Nu
180 prop_binCROS_muon2017_bin9
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

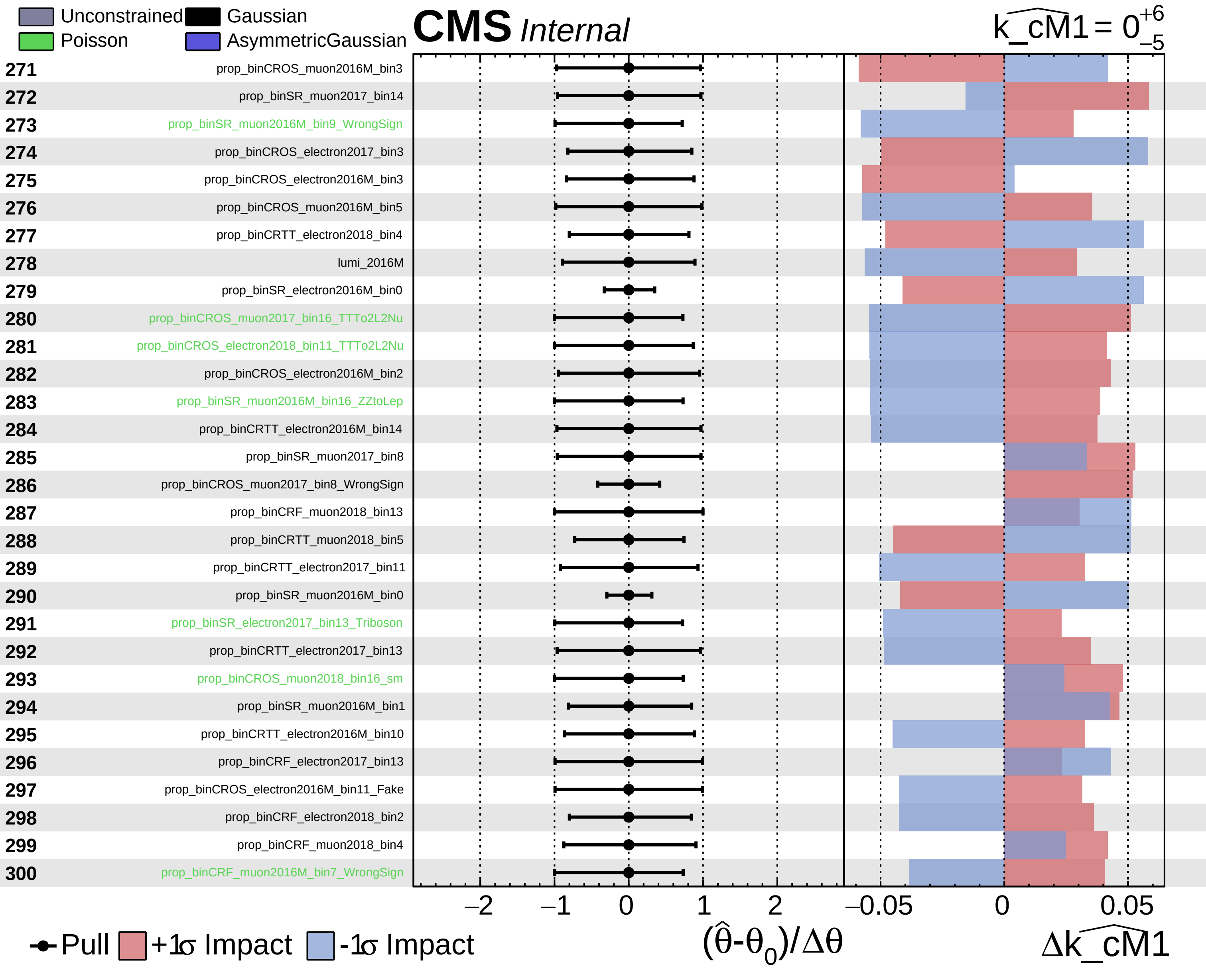


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
181 prop_binCROS_electron2018_bin7_WrongSign
182 prop_binCROS_muon2018_bin13
183 prop_binCRTT_electron2017_bin1
184 prop_binCROS_electron2017_bin11_WrongSign
185 prop_binSR_muon2018_bin3
186 prop_binCRTT_electron2018_bin14
187 prop_binCRF_electron2018_bin4
188 prop_binSR_electron2018_bin2
189 prop_binSR_muon2018_bin4
190 prop_binCROS_muon2016M_bin13
191 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
192 prop_binCRF_electron2018_bin6
193 prop_binSR_electron2017_bin3
194 prop_binCROS_electron2018_bin9
195 prop_binSR_muon2017_bin2
196 prop_binCRTT_electron2018_bin1
197 prop_binCRF_electron2017_bin3
198 prop_binCRF_muon2018_bin12
199 prop_binSR_electron2018_bin15
200 prop_binCROS_electron2016M_bin9
201 prop_binCROS_electron2016M_bin13_Fake
202 prop_binCRF_muon2018_bin2
203 prop_binCROS_electron2017_bin8
204 prop_binCROS_muon2016M_bin14_TTTo2L2Nu
205 prop_binCRTT_muon2016M_bin1
206 prop_binCROS_muon2018_bin3
207 prop_binSR_muon2018_bin12_WrongSign
208 prop_binCROS_muon2017_bin12_WrongSign
209 prop_binCRF_muon2016M_bin7_sm
210 prop_binSR_electron2018_bin8
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

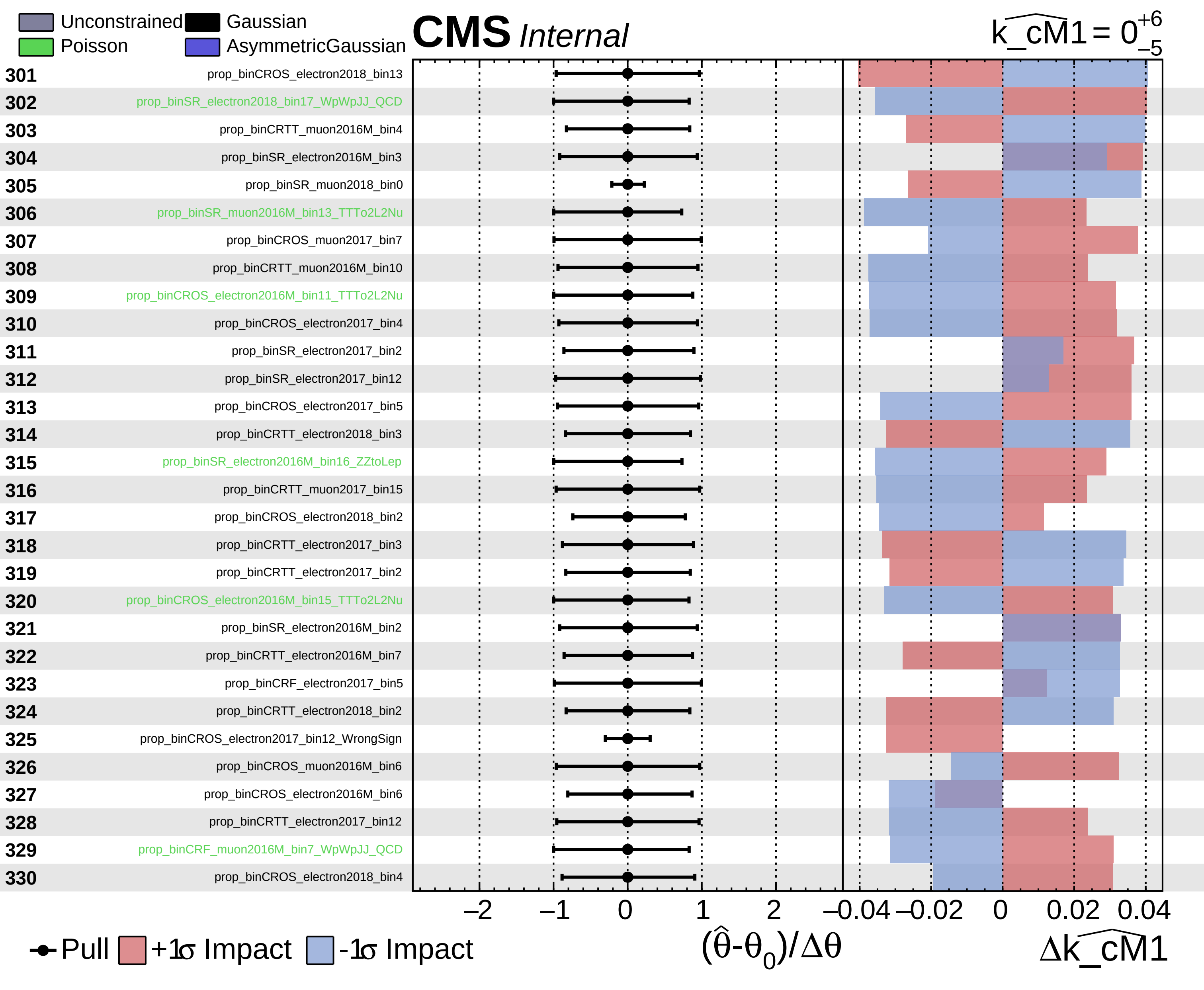

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
211 prop_binCRTT_electron2016M_bin1
212 prop_binCROS_muon2016M_bin9
213 prop_binCROS_muon2016M_bin11
214 prop_binCROS_electron2016M_bin13_WrongSign
215 prop_binSR_electron2017_bin1
216 prop_binCROS_electron2018_bin6
217 prop_binCRF_electron2017_bin4_TTTo2L2Nu
218 prop_binSR_muon2016M_bin4
219 prop_binSR_muon2018_bin1
220 prop_binCROS_electron2018_bin3
221 prop_binCRF_muon2017_bin0
222 prop_binCRTT_muon2018_bin3
223 prop_binCROS_electron2018_bin12
224 prop_binCROS_electron2017_bin1
225 prop_binCRF_muon2017_bin3
226 prop_binSR_muon2017_bin1
227 prop_binSR_electron2018_bin1
228 tau_vsjet_2018
229 prop_binCROS_muon2016M_bin10
230 prop_binCROS_electron2018_bin14_WrongSign
231 tau_vsjet_2017
232 prop_binCRF_muon2018_bin6
233 prop_binCROS_electron2017_bin2
234 prop_binSR_muon2016M_bin14_TTTo2L2Nu
235 prop_binSR_electron2017_bin14_TTTo2L2Nu
236 prop_binCRF_muon2017_bin12
237 prop_binCROS_muon2017_bin5_WrongSign
238 prop_binCRTT_muon2018_bin1
239 prop_binSR_electron2017_bin5
240 prop_binCRTT_muon2016M_bin16
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1

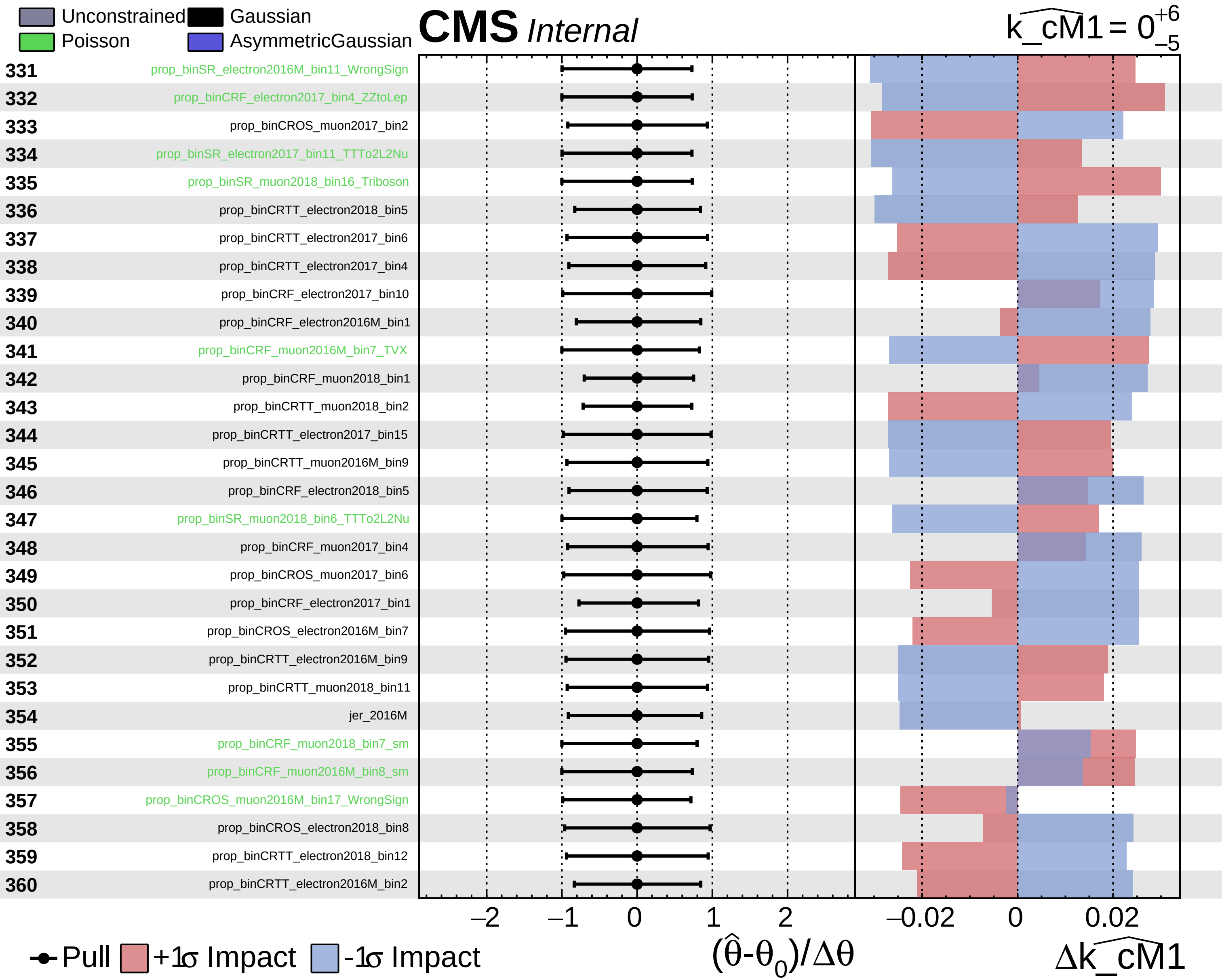


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0+6−5
241 prop_binSR_electron2016M_bin6
242 prop_binCROS_electron2018_bin5
243 prop_binSR_electron2018_bin5
244 prop_binCROS_electron2018_bin10
245 lumi_2017
246 prop_binCROS_electron2016M_bin8
247 prop_binSR_muon2016M_bin15
248 prop_binCROS_electron2016M_bin16_WrongSign
249 prop_binSR_electron2018_bin17_Triboson
250 prop_binSR_electron2018_bin4
251 prop_binCROS_muon2018_bin8
252 prop_binCROS_electron2017_bin7
253 prop_binCROS_muon2018_bin7
254 prop_binCRF_electron2018_bin3
255 prop_binCRF_electron2017_bin11_sm
256 prop_binSR_muon2017_bin3
257 prop_binSR_muon2017_bin0
258 prop_binCRF_electron2017_bin4_sm
259 prop_binCRTT_muon2017_bin1
260 prop_binSR_electron2017_bin0
261 prop_binCROS_electron2017_bin11_TTTo2L2Nu
262 prop_binCRF_muon2018_bin3
263 prop_binSR_electron2016M_bin4
264 prop_binSR_electron2016M_bin1
265 prop_binSR_muon2016M_bin12_WrongSign
266 prop_binCROS_electron2016M_bin4
267 prop_binSR_electron2018_bin11_VG
268 prop_binCROS_muon2017_bin16_TVX
269 prop_binCRF_electron2016M_bin0
270 prop_binCRF_muon2016M_bin13
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1

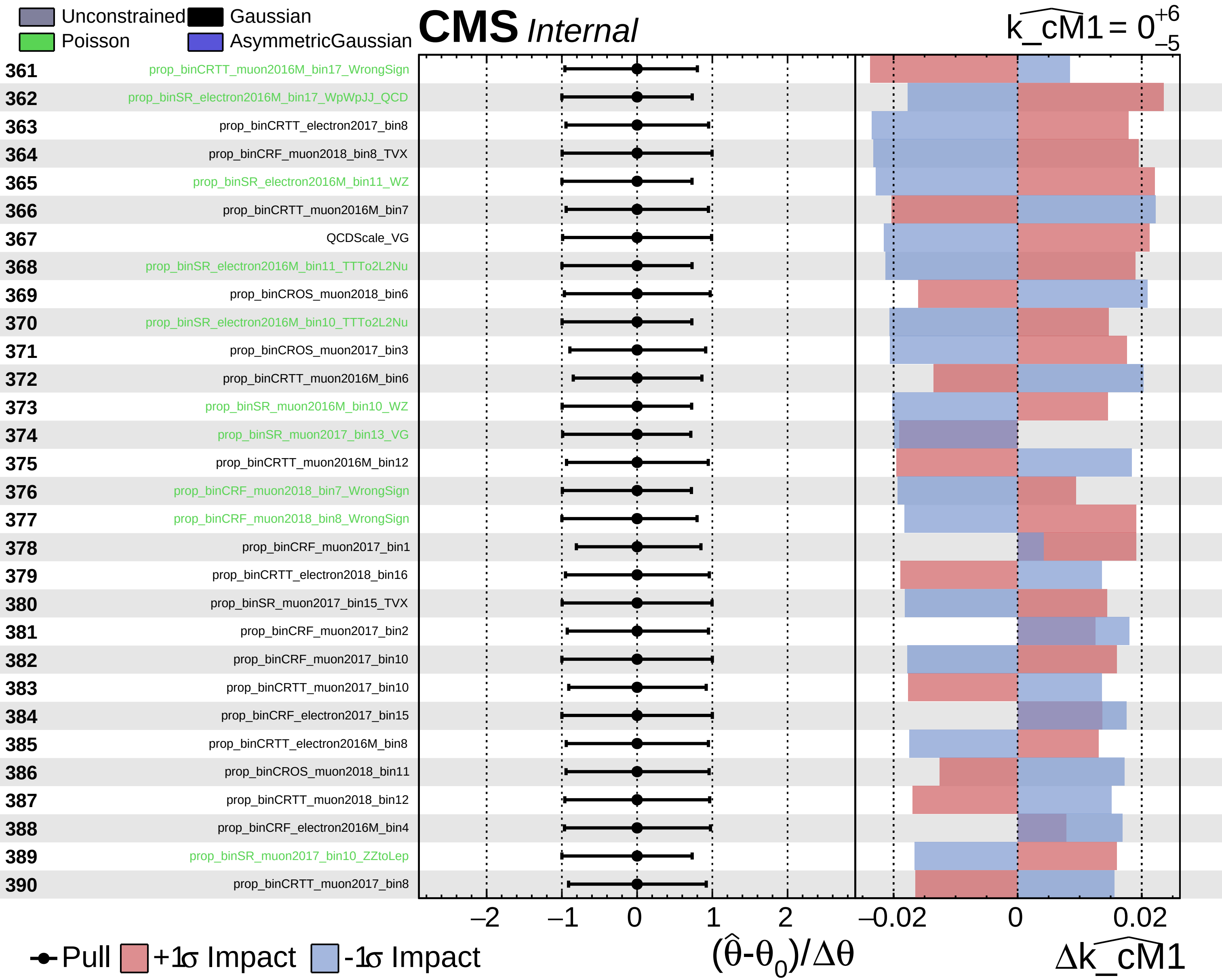


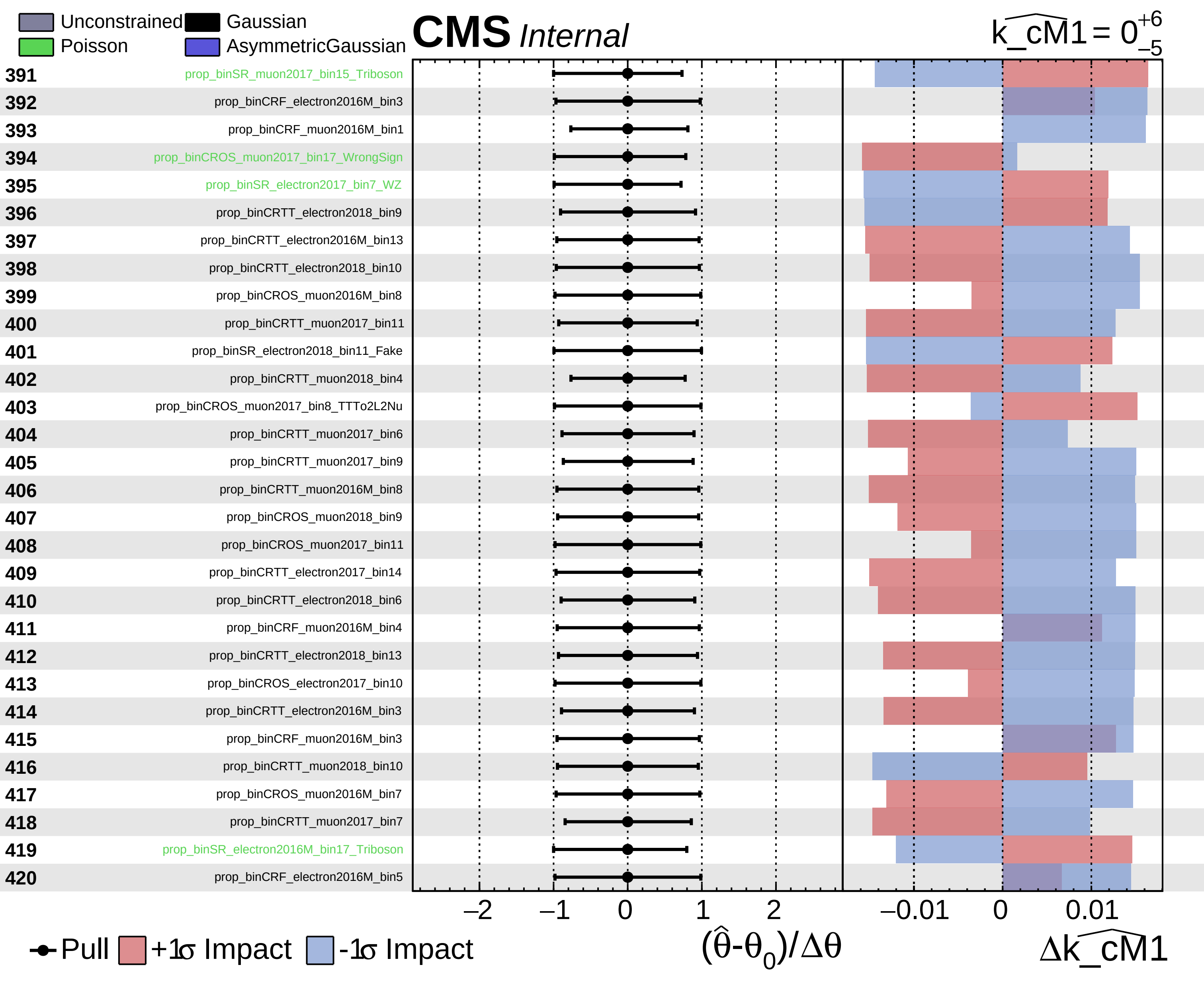
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
271 prop_binCROS_muon2016M_bin3
272 prop_binSR_muon2017_bin14
273 prop_binSR_muon2016M_bin9_WrongSign
274 prop_binCROS_electron2017_bin3
275 prop_binCROS_electron2016M_bin3
276 prop_binCROS_muon2016M_bin5
277 prop_binCRTT_electron2018_bin4
278 lumi_2016M
279 prop_binSR_electron2016M_bin0
280 prop_binCROS_muon2017_bin16_TTTo2L2Nu
281 prop_binCROS_electron2018_bin11_TTTo2L2Nu
282 prop_binCROS_electron2016M_bin2
283 prop_binSR_muon2016M_bin16_ZZtoLep
284 prop_binCRTT_electron2016M_bin14
285 prop_binSR_muon2017_bin8
286 prop_binCROS_muon2017_bin8_WrongSign
287 prop_binCRF_muon2018_bin13
288 prop_binCRTT_muon2018_bin5
289 prop_binCRTT_electron2017_bin11
290 prop_binSR_muon2016M_bin0
291 prop_binSR_electron2017_bin13_Triboson
292 prop_binCRTT_electron2017_bin13
293 prop_binCROS_muon2018_bin16_sm
294 prop_binSR_muon2016M_bin1
295 prop_binCRTT_electron2016M_bin10
296 prop_binCRF_electron2017_bin13
297 prop_binCROS_electron2016M_bin11_Fake
298 prop_binCRF_electron2018_bin2
299 prop_binCRF_muon2018_bin4
300 prop_binCRF_muon2016M_bin7_WrongSign
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
301 prop_binCROS_electron2018_bin13
302 prop_binSR_electron2018_bin17_WpWpJJ_QCD
303 prop_binCRTT_muon2016M_bin4
304 prop_binSR_electron2016M_bin3
305 prop_binSR_muon2018_bin0
306 prop_binSR_muon2016M_bin13_TTTo2L2Nu
307 prop_binCROS_muon2017_bin7
308 prop_binCRTT_muon2016M_bin10
309 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
310 prop_binCROS_electron2017_bin4
311 prop_binSR_electron2017_bin2
312 prop_binSR_electron2017_bin12
313 prop_binCROS_electron2017_bin5
314 prop_binCRTT_electron2018_bin3
315 prop_binSR_electron2016M_bin16_ZZtoLep
316 prop_binCRTT_muon2017_bin15
317 prop_binCROS_electron2018_bin2
318 prop_binCRTT_electron2017_bin3
319 prop_binCRTT_electron2017_bin2
320 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
321 prop_binSR_electron2016M_bin2
322 prop_binCRTT_electron2016M_bin7
323 prop_binCRF_electron2017_bin5
324 prop_binCRTT_electron2018_bin2
325 prop_binCROS_electron2017_bin12_WrongSign
326 prop_binCROS_muon2016M_bin6
327 prop_binCROS_electron2016M_bin6
328 prop_binCRTT_electron2017_bin12
329 prop_binCRF_muon2016M_bin7_WpWpJJ_QCD
330 prop_binCROS_electron2018_bin4
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1

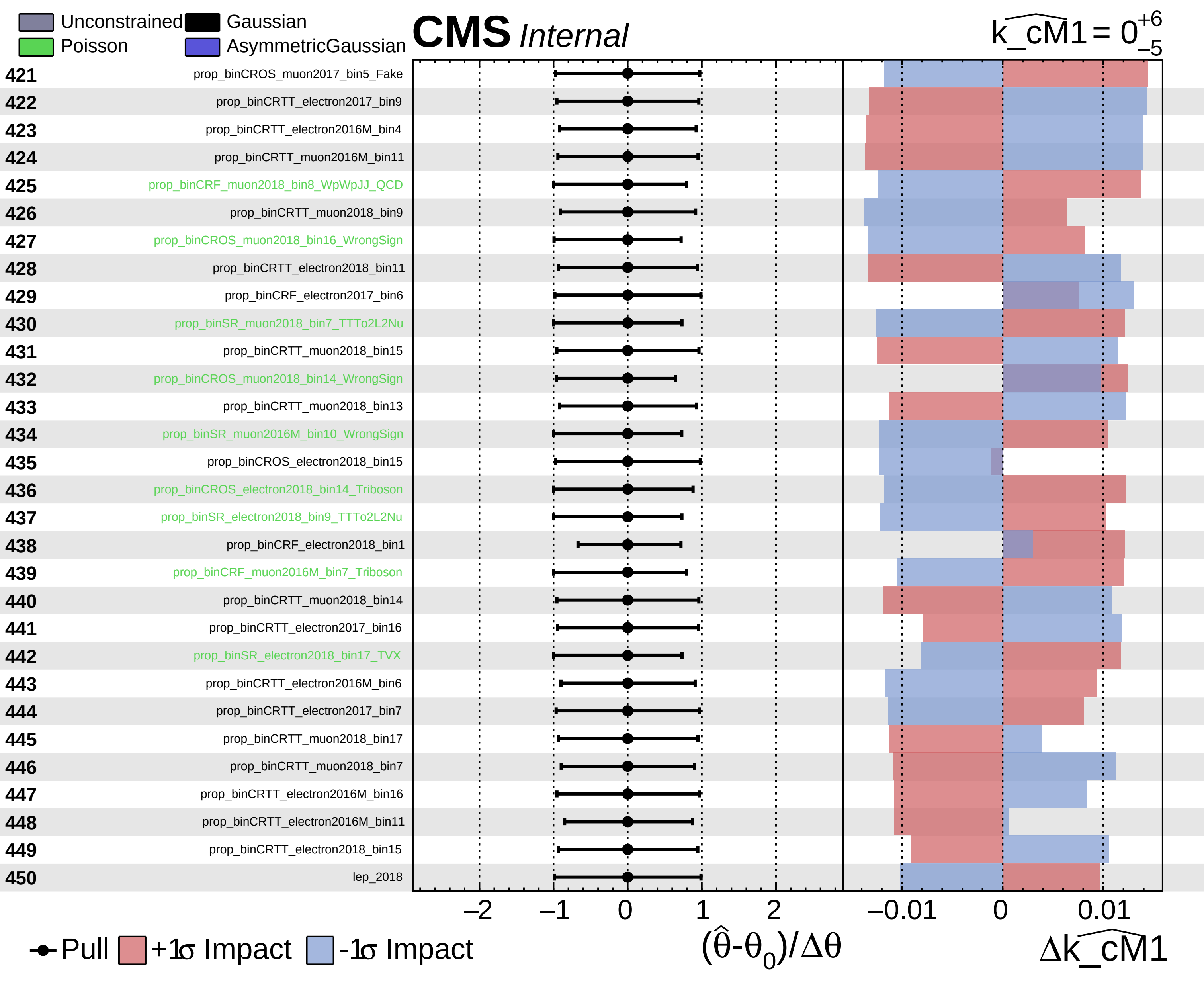


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
331 prop_binSR_electron2016M_bin11_WrongSign
332 prop_binCRF_electron2017_bin4_ZZtoLep
333 prop_binCROS_muon2017_bin2
334 prop_binSR_electron2017_bin11_TTTo2L2Nu
335 prop_binSR_muon2018_bin16_Triboson
336 prop_binCRTT_electron2018_bin5
337 prop_binCRTT_electron2017_bin6
338 prop_binCRTT_electron2017_bin4
339 prop_binCRF_electron2017_bin10
340 prop_binCRF_electron2016M_bin1
341 prop_binCRF_muon2016M_bin7_TVX
342 prop_binCRF_muon2018_bin1
343 prop_binCRTT_muon2018_bin2
344 prop_binCRTT_electron2017_bin15
345 prop_binCRTT_muon2016M_bin9
346 prop_binCRF_electron2018_bin5
347 prop_binSR_muon2018_bin6_TTTo2L2Nu
348 prop_binCRF_muon2017_bin4
349 prop_binCROS_muon2017_bin6
350 prop_binCRF_electron2017_bin1
351 prop_binCROS_electron2016M_bin7
352 prop_binCRTT_electron2016M_bin9
353 prop_binCRTT_muon2018_bin11
354 jer_2016M
355 prop_binCRF_muon2018_bin7_sm
356 prop_binCRF_muon2016M_bin8_sm
357 prop_binCROS_muon2016M_bin17_WrongSign
358 prop_binCROS_electron2018_bin8
359 prop_binCRTT_electron2018_bin12
360 prop_binCRTT_electron2016M_bin2
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

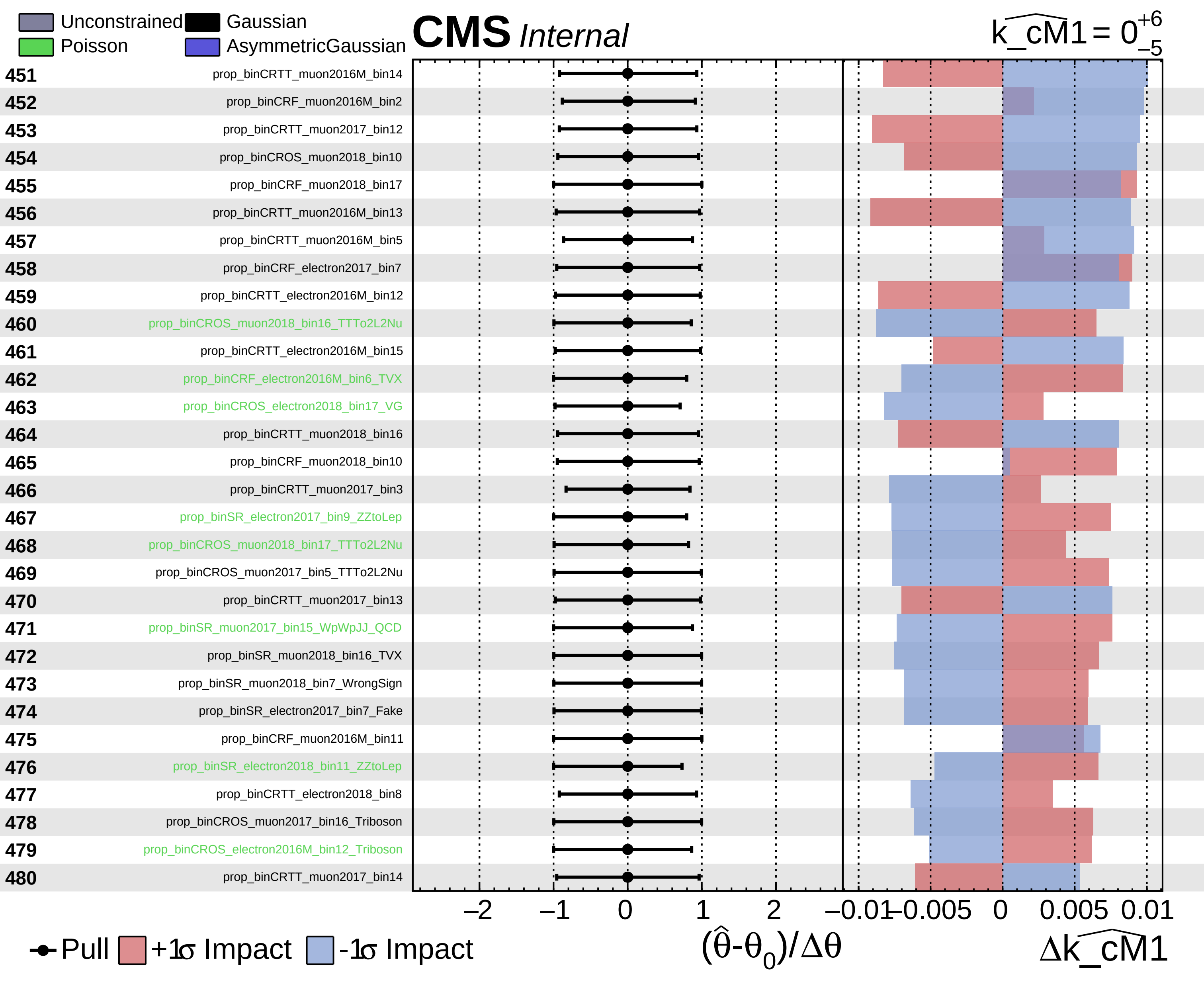

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
361 prop_binCRTT_muon2016M_bin17_WrongSign
362 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
363 prop_binCRTT_electron2017_bin8
364 prop_binCRF_muon2018_bin8_TVX
365 prop_binSR_electron2016M_bin11_WZ
366 prop_binCRTT_muon2016M_bin7
367 QCDScale_VG
368 prop_binSR_electron2016M_bin11_TTTo2L2Nu
369 prop_binCROS_muon2018_bin6
370 prop_binSR_electron2016M_bin10_TTTo2L2Nu
371 prop_binCROS_muon2017_bin3
372 prop_binCRTT_muon2016M_bin6
373 prop_binSR_muon2016M_bin10_WZ
374 prop_binSR_muon2017_bin13_VG
375 prop_binCRTT_muon2016M_bin12
376 prop_binCRF_muon2018_bin7_WrongSign
377 prop_binCRF_muon2018_bin8_WrongSign
378 prop_binCRF_muon2017_bin1
379 prop_binCRTT_electron2018_bin16
380 prop_binSR_muon2017_bin15_TVX
381 prop_binCRF_muon2017_bin2
382 prop_binCRF_muon2017_bin10
383 prop_binCRTT_muon2017_bin10
384 prop_binCRF_electron2017_bin15
385 prop_binCRTT_electron2016M_bin8
386 prop_binCROS_muon2018_bin11
387 prop_binCRTT_muon2018_bin12
388 prop_binCRF_electron2016M_bin4
389 prop_binSR_muon2017_bin10_ZZtoLep
390 prop_binCRTT_muon2017_bin8
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



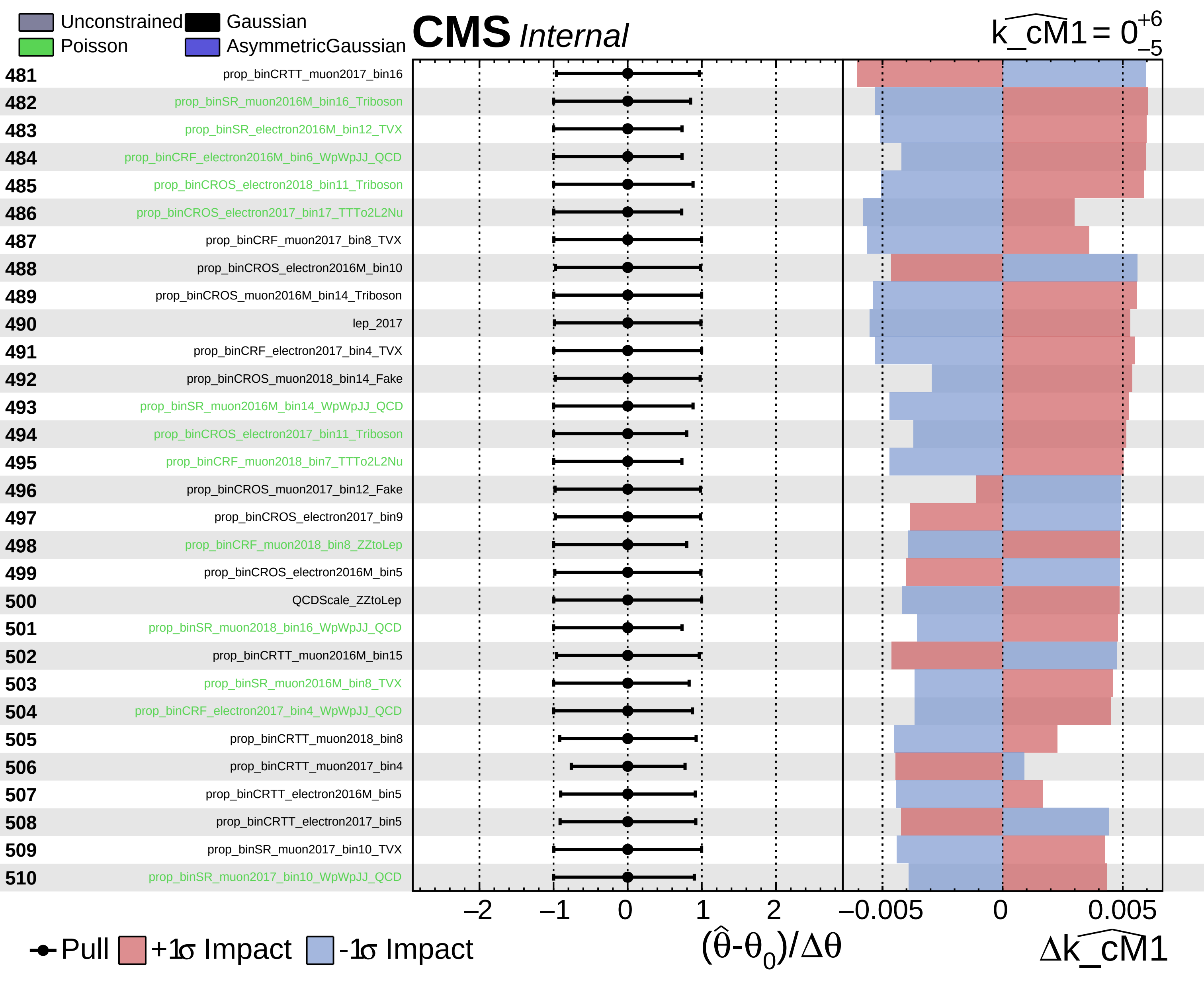
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
391 prop_binSR_muon2017_bin15_Triboson
392 prop_binCRF_electron2016M_bin3
393 prop_binCRF_muon2016M_bin1
394 prop_binCROS_muon2017_bin17_WrongSign
395 prop_binSR_electron2017_bin7_WZ
396 prop_binCRTT_electron2018_bin9
397 prop_binCRTT_electron2016M_bin13
398 prop_binCRTT_electron2018_bin10
399 prop_binCROS_muon2016M_bin8
400 prop_binCRTT_muon2017_bin11
401 prop_binSR_electron2018_bin11_Fake
402 prop_binCRTT_muon2018_bin4
403 prop_binCROS_muon2017_bin8_TTTo2L2Nu
404 prop_binCRTT_muon2017_bin6
405 prop_binCRTT_muon2017_bin9
406 prop_binCRTT_muon2016M_bin8
407 prop_binCROS_muon2018_bin9
408 prop_binCROS_muon2017_bin11
409 prop_binCRTT_electron2017_bin14
410 prop_binCRTT_electron2018_bin6
411 prop_binCRF_muon2016M_bin4
412 prop_binCRTT_electron2018_bin13
413 prop_binCROS_electron2017_bin10
414 prop_binCRTT_electron2016M_bin3
415 prop_binCRF_muon2016M_bin3
416 prop_binCRTT_muon2018_bin10
417 prop_binCROS_muon2016M_bin7
418 prop_binCRTT_muon2017_bin7
419 prop_binSR_electron2016M_bin17_Triboson
420 prop_binCRF_electron2016M_bin5
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



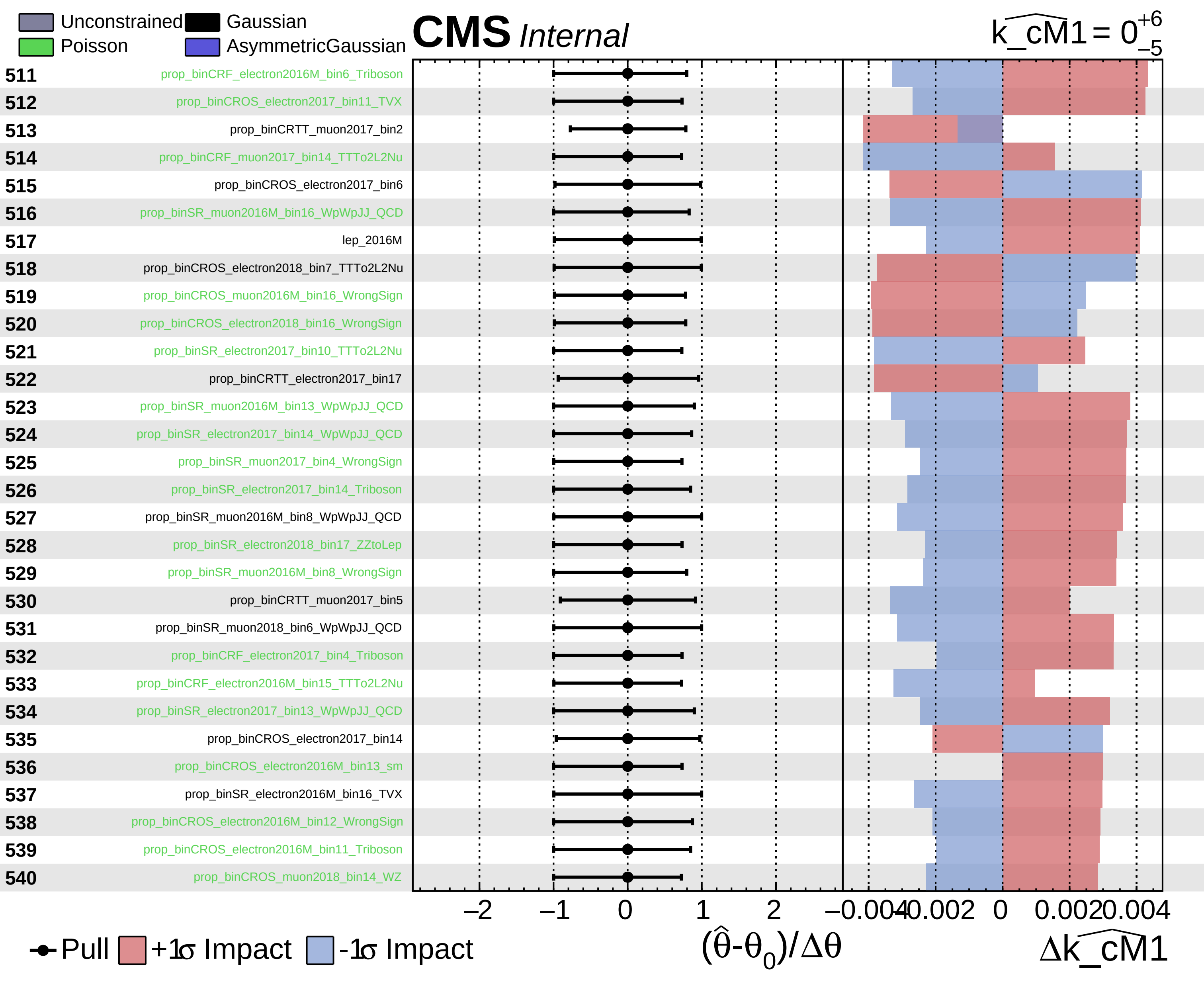
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
421 prop_binCROS_muon2017_bin5_Fake
422 prop_binCRTT_electron2017_bin9
423 prop_binCRTT_electron2016M_bin4
424 prop_binCRTT_muon2016M_bin11
425 prop_binCRF_muon2018_bin8_WpWpJJ_QCD
426 prop_binCRTT_muon2018_bin9
427 prop_binCROS_muon2018_bin16_WrongSign
428 prop_binCRTT_electron2018_bin11
429 prop_binCRF_electron2017_bin6
430 prop_binSR_muon2018_bin7_TTTo2L2Nu
431 prop_binCRTT_muon2018_bin15
432 prop_binCROS_muon2018_bin14_WrongSign
433 prop_binCRTT_muon2018_bin13
434 prop_binSR_muon2016M_bin10_WrongSign
435 prop_binCROS_electron2018_bin15
436 prop_binCROS_electron2018_bin14_Triboson
437 prop_binSR_electron2018_bin9_TTTo2L2Nu
438 prop_binCRF_electron2018_bin1
439 prop_binCRF_muon2016M_bin7_Triboson
440 prop_binCRTT_muon2018_bin14
441 prop_binCRTT_electron2017_bin16
442 prop_binSR_electron2018_bin17_TVX
443 prop_binCRTT_electron2016M_bin6
444 prop_binCRTT_electron2017_bin7
445 prop_binCRTT_muon2018_bin17
446 prop_binCRTT_muon2018_bin7
447 prop_binCRTT_electron2016M_bin16
448 prop_binCRTT_electron2016M_bin11
449 prop_binCRTT_electron2018_bin15
450 lep_2018
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1



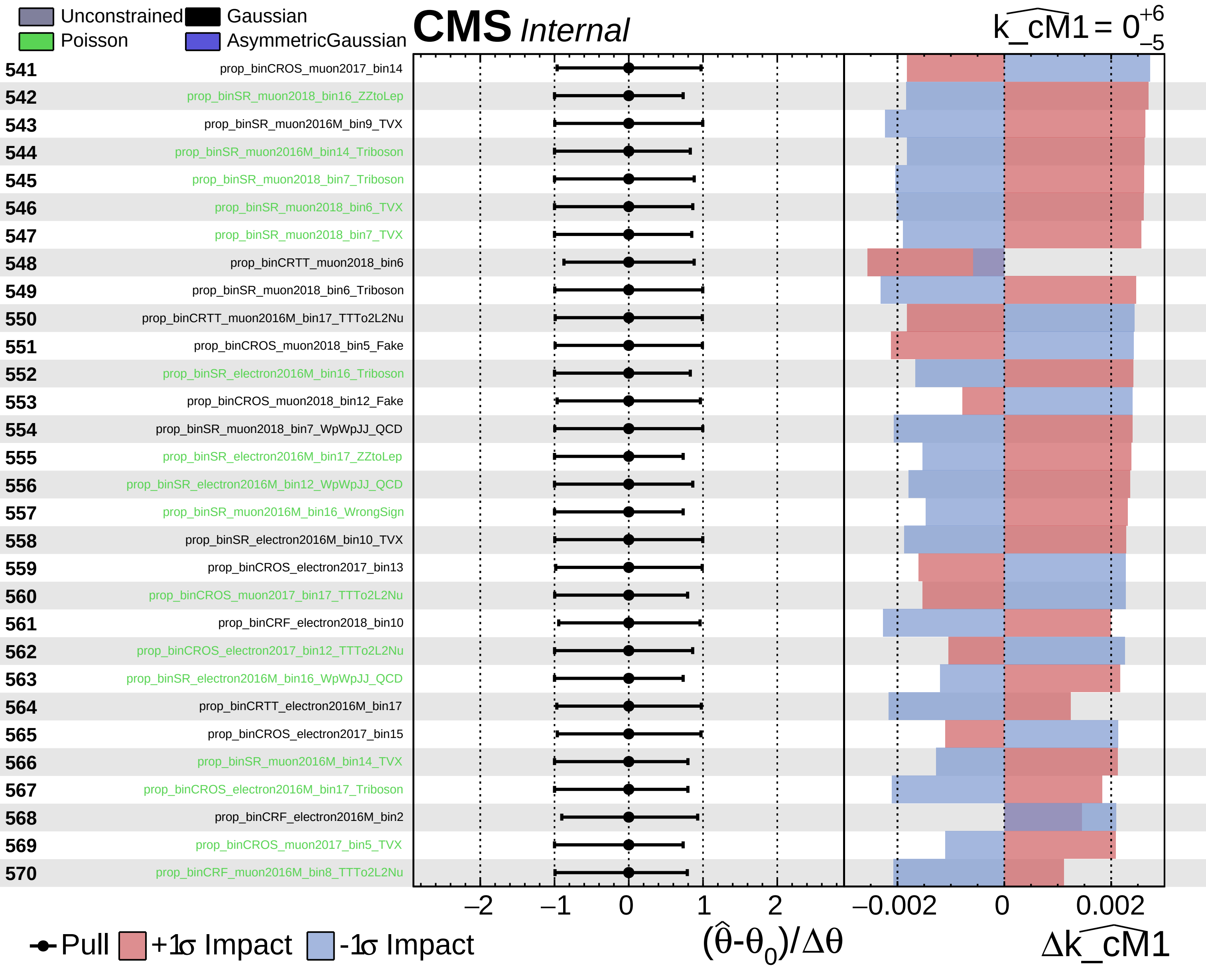
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
451 prop_binCRTT_muon2016M_bin14
452 prop_binCRF_muon2016M_bin2
453 prop_binCRTT_muon2017_bin12
454 prop_binCROS_muon2018_bin10
455 prop_binCRF_muon2018_bin17
456 prop_binCRTT_muon2016M_bin13
457 prop_binCRTT_muon2016M_bin5
458 prop_binCRF_electron2017_bin7
459 prop_binCRTT_electron2016M_bin12
460 prop_binCROS_muon2018_bin16_TTTo2L2Nu
461 prop_binCRTT_electron2016M_bin15
462 prop_binCRF_electron2016M_bin6_TVX
463 prop_binCROS_electron2018_bin17_VG
464 prop_binCRTT_muon2018_bin16
465 prop_binCRF_muon2018_bin10
466 prop_binCRTT_muon2017_bin3
467 prop_binSR_electron2017_bin9_ZZtoLep
468 prop_binCROS_muon2018_bin17_TTTo2L2Nu
469 prop_binCROS_muon2017_bin5_TTTo2L2Nu
470 prop_binCRTT_muon2017_bin13
471 prop_binSR_muon2017_bin15_WpWpJJ_QCD
472 prop_binSR_muon2018_bin16_TVX
473 prop_binSR_muon2018_bin7_WrongSign
474 prop_binSR_electron2017_bin7_Fake
475 prop_binCRF_muon2016M_bin11
476 prop_binSR_electron2018_bin11_ZZtoLep
477 prop_binCRTT_electron2018_bin8
478 prop_binCROS_muon2017_bin16_Triboson
479 prop_binCROS_electron2016M_bin12_Triboson
480 prop_binCRTT_muon2017_bin14
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1



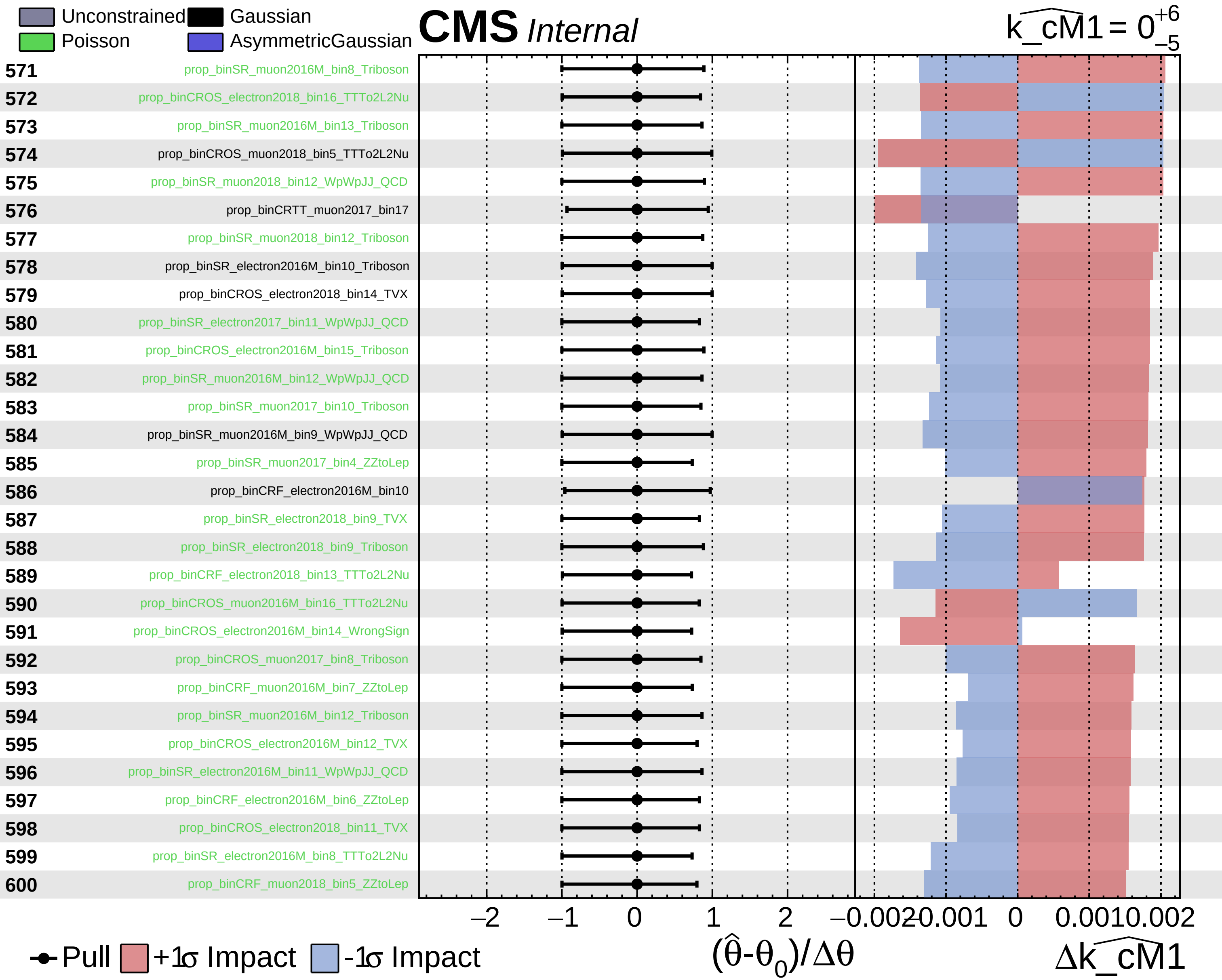
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
481 prop_binCRTT_muon2017_bin16
482 prop_binSR_muon2016M_bin16_Triboson
483 prop_binSR_electron2016M_bin12_TVX
484 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
485 prop_binCROS_electron2018_bin11_Triboson
486 prop_binCROS_electron2017_bin17_TTTo2L2Nu
487 prop_binCRF_muon2017_bin8_TVX
488 prop_binCROS_electron2016M_bin10
489 prop_binCROS_muon2016M_bin14_Triboson
490 lep_2017
491 prop_binCRF_electron2017_bin4_TVX
492 prop_binCROS_muon2018_bin14_Fake
493 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
494 prop_binCROS_electron2017_bin11_Triboson
495 prop_binCRF_muon2018_bin7_TTTo2L2Nu
496 prop_binCROS_muon2017_bin12_Fake
497 prop_binCROS_electron2017_bin9
498 prop_binCRF_muon2018_bin8_ZZtoLep
499 prop_binCROS_electron2016M_bin5
500 QCDScale_ZZtoLep
501 prop_binSR_muon2018_bin16_WpWpJJ_QCD
502 prop_binCRTT_muon2016M_bin15
503 prop_binSR_muon2016M_bin8_TVX
504 prop_binCRF_electron2017_bin4_WpWpJJ_QCD
505 prop_binCRTT_muon2018_bin8
506 prop_binCRTT_muon2017_bin4
507 prop_binCRTT_electron2016M_bin5
508 prop_binCRTT_electron2017_bin5
509 prop_binSR_muon2017_bin10_TVX
510 prop_binSR_muon2017_bin10_WpWpJJ_QCD
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



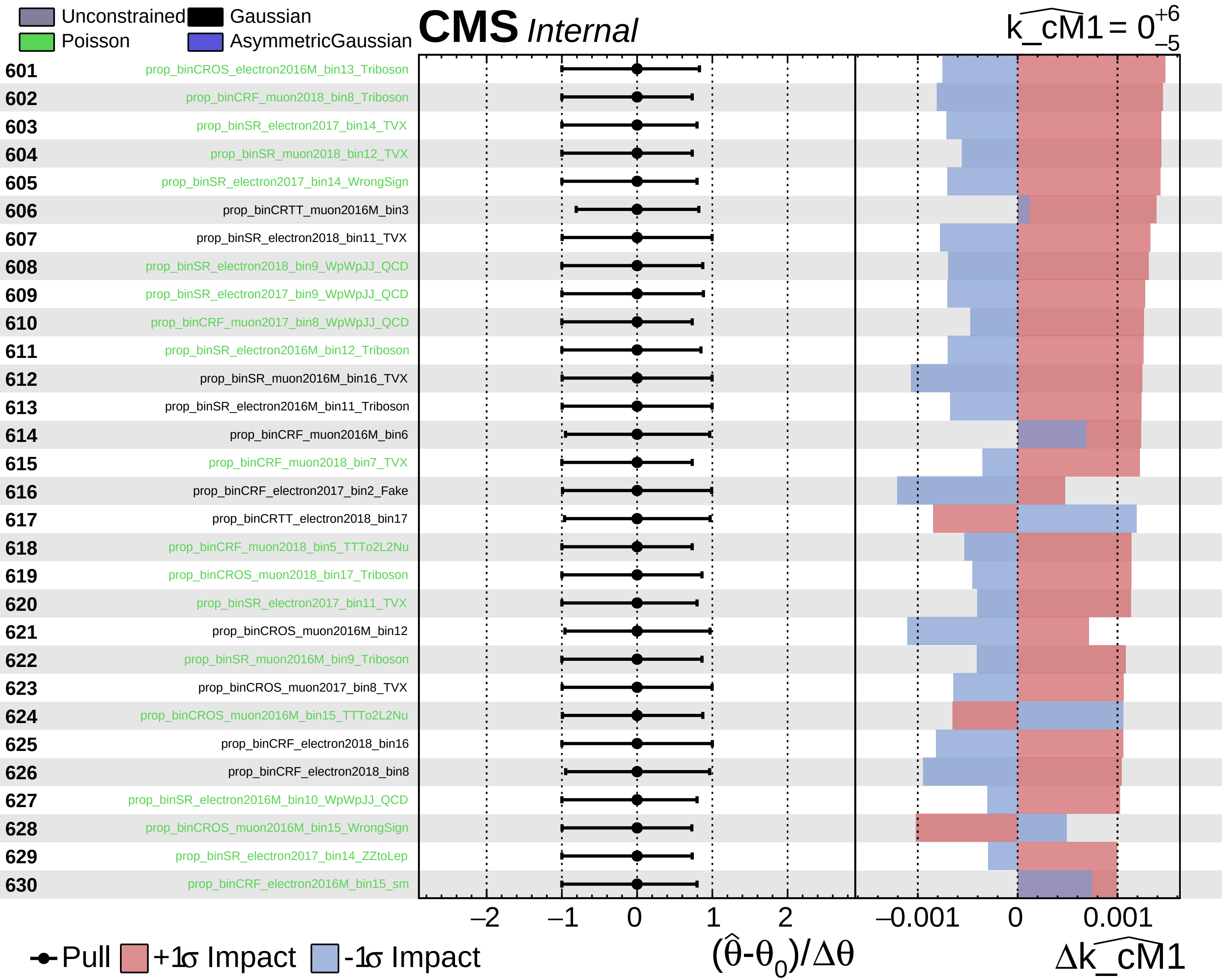
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0+6−5
511 prop_binCRF_electron2016M_bin6_Triboson
512 prop_binCROS_electron2017_bin11_TVX
513 prop_binCRTT_muon2017_bin2
514 prop_binCRF_muon2017_bin14_TTTo2L2Nu
515 prop_binCROS_electron2017_bin6
516 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
517 lep_2016M
518 prop_binCROS_electron2018_bin7_TTTo2L2Nu
519 prop_binCROS_muon2016M_bin16_WrongSign
520 prop_binCROS_electron2018_bin16_WrongSign
521 prop_binSR_electron2017_bin10_TTTo2L2Nu
522 prop_binCRTT_electron2017_bin17
523 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
524 prop_binSR_electron2017_bin14_WpWpJJ_QCD
525 prop_binSR_muon2017_bin4_WrongSign
526 prop_binSR_electron2017_bin14_Triboson
527 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
528 prop_binSR_electron2018_bin17_ZZtoLep
529 prop_binSR_muon2016M_bin8_WrongSign
530 prop_binCRTT_muon2017_bin5
531 prop_binSR_muon2018_bin6_WpWpJJ_QCD
532 prop_binCRF_electron2017_bin4_Triboson
533 prop_binCRF_electron2016M_bin15_TTTo2L2Nu
534 prop_binSR_electron2017_bin13_WpWpJJ_QCD
535 prop_binCROS_electron2017_bin14
536 prop_binCROS_electron2016M_bin13_sm
537 prop_binSR_electron2016M_bin16_TVX
538 prop_binCROS_electron2016M_bin12_WrongSign
539 prop_binCROS_electron2016M_bin11_Triboson
540 prop_binCROS_muon2018_bin14_WZ
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1



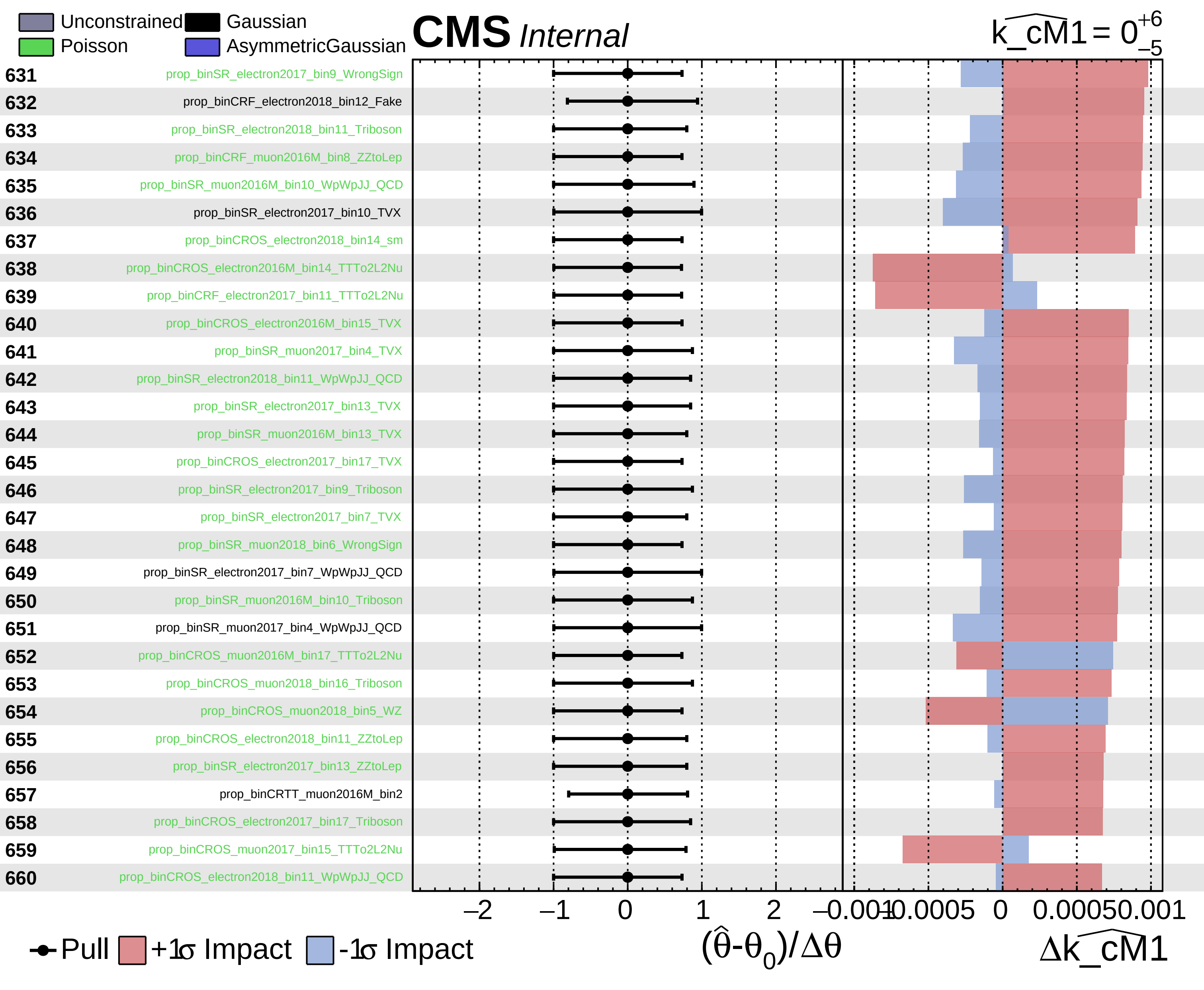
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
541 prop_binCROS_muon2017_bin14
542 prop_binSR_muon2018_bin16_ZZtoLep
543 prop_binSR_muon2016M_bin9_TVX
544 prop_binSR_muon2016M_bin14_Triboson
545 prop_binSR_muon2018_bin7_Triboson
546 prop_binSR_muon2018_bin6_TVX
547 prop_binSR_muon2018_bin7_TVX
548 prop_binCRTT_muon2018_bin6
549 prop_binSR_muon2018_bin6_Triboson
550 prop_binCRTT_muon2016M_bin17_TTTo2L2Nu
551 prop_binCROS_muon2018_bin5_Fake
552 prop_binSR_electron2016M_bin16_Triboson
553 prop_binCROS_muon2018_bin12_Fake
554 prop_binSR_muon2018_bin7_WpWpJJ_QCD
555 prop_binSR_electron2016M_bin17_ZZtoLep
556 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
557 prop_binSR_muon2016M_bin16_WrongSign
558 prop_binSR_electron2016M_bin10_TVX
559 prop_binCROS_electron2017_bin13
560 prop_binCROS_muon2017_bin17_TTTo2L2Nu
561 prop_binCRF_electron2018_bin10
562 prop_binCROS_electron2017_bin12_TTTo2L2Nu
563 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
564 prop_binCRTT_electron2016M_bin17
565 prop_binCROS_electron2017_bin15
566 prop_binSR_muon2016M_bin14_TVX
567 prop_binCROS_electron2016M_bin17_Triboson
568 prop_binCRF_electron2016M_bin2
569 prop_binCROS_muon2017_bin5_TVX
570 prop_binCRF_muon2016M_bin8_TTTo2L2Nu
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



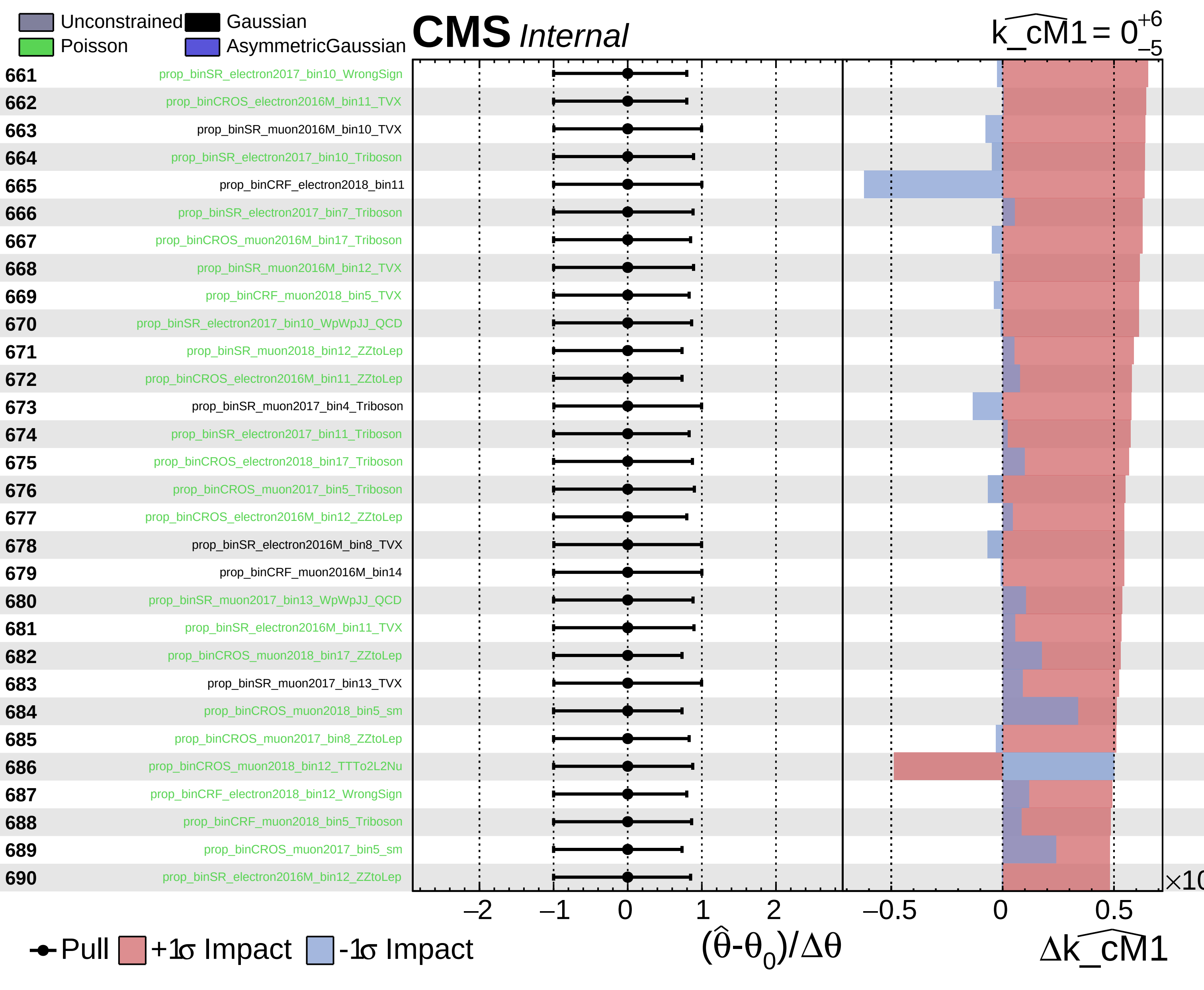
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
571 prop_binSR_muon2016M_bin8_Triboson
572 prop_binCROS_electron2018_bin16_TTTo2L2Nu
573 prop_binSR_muon2016M_bin13_Triboson
574 prop_binCROS_muon2018_bin5_TTTo2L2Nu
575 prop_binSR_muon2018_bin12_WpWpJJ_QCD
576 prop_binCRTT_muon2017_bin17
577 prop_binSR_muon2018_bin12_Triboson
578 prop_binSR_electron2016M_bin10_Triboson
579 prop_binCROS_electron2018_bin14_TVX
580 prop_binSR_electron2017_bin11_WpWpJJ_QCD
581 prop_binCROS_electron2016M_bin15_Triboson
582 prop_binSR_muon2016M_bin12_WpWpJJ_QCD
583 prop_binSR_muon2017_bin10_Triboson
584 prop_binSR_muon2016M_bin9_WpWpJJ_QCD
585 prop_binSR_muon2017_bin4_ZZtoLep
586 prop_binCRF_electron2016M_bin10
587 prop_binSR_electron2018_bin9_TVX
588 prop_binSR_electron2018_bin9_Triboson
589 prop_binCRF_electron2018_bin13_TTTo2L2Nu
590 prop_binCROS_muon2016M_bin16_TTTo2L2Nu
591 prop_binCROS_electron2016M_bin14_WrongSign
592 prop_binCROS_muon2017_bin8_Triboson
593 prop_binCRF_muon2016M_bin7_ZZtoLep
594 prop_binSR_muon2016M_bin12_Triboson
595 prop_binCROS_electron2016M_bin12_TVX
596 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
597 prop_binCRF_electron2016M_bin6_ZZtoLep
598 prop_binCROS_electron2018_bin11_TVX
599 prop_binSR_electron2016M_bin8_TTTo2L2Nu
600 prop_binCRF_muon2018_bin5_ZZtoLep
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



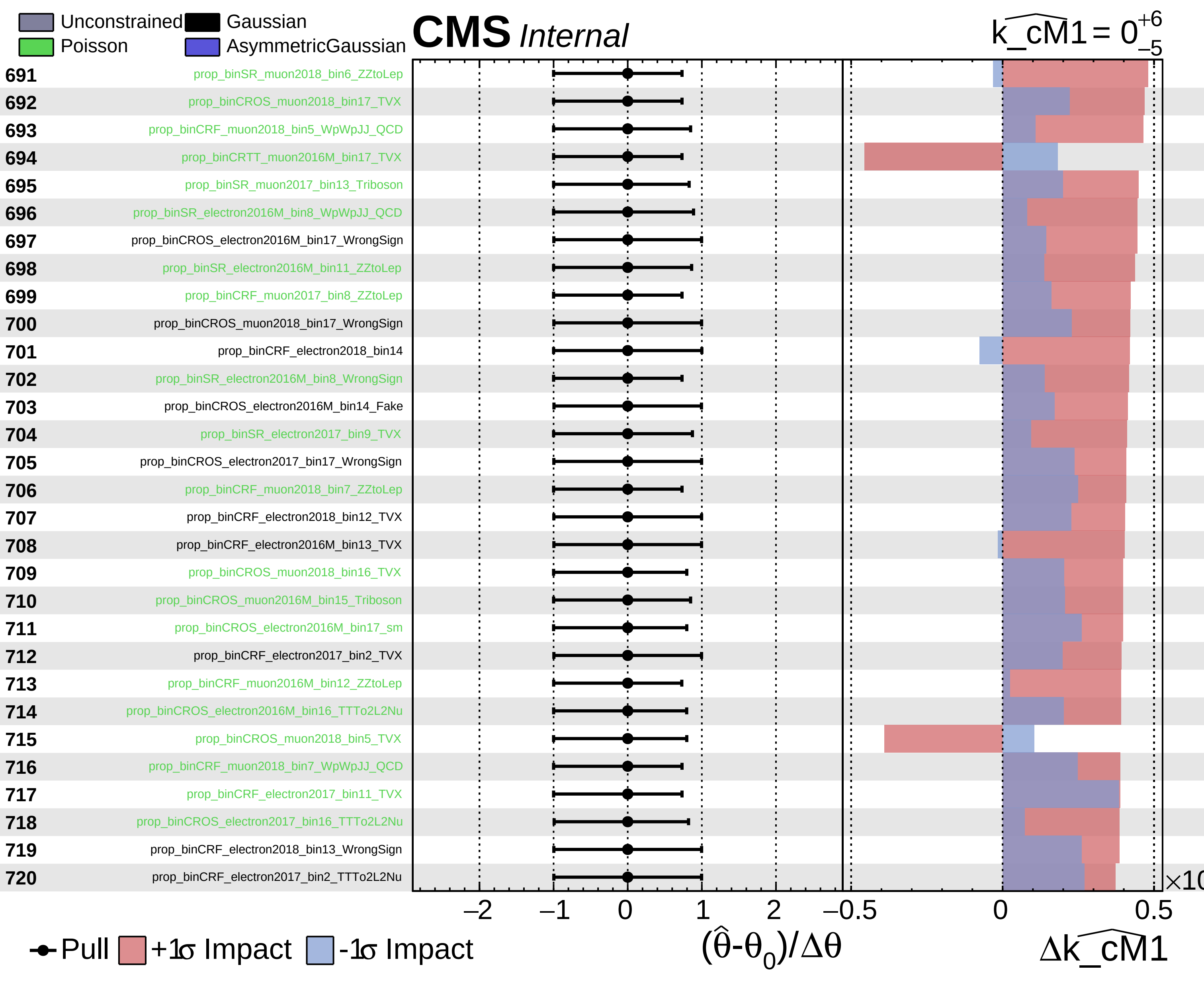
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
601 prop_binCROS_electron2016M_bin13_Triboson
602 prop_binCRF_muon2018_bin8_Triboson
603 prop_binSR_electron2017_bin14_TVX
604 prop_binSR_muon2018_bin12_TVX
605 prop_binSR_electron2017_bin14_WrongSign
606 prop_binCRTT_muon2016M_bin3
607 prop_binSR_electron2018_bin11_TVX
608 prop_binSR_electron2018_bin9_WpWpJJ_QCD
609 prop_binSR_electron2017_bin9_WpWpJJ_QCD
610 prop_binCRF_muon2017_bin8_WpWpJJ_QCD
611 prop_binSR_electron2016M_bin12_Triboson
612 prop_binSR_muon2016M_bin16_TVX
613 prop_binSR_electron2016M_bin11_Triboson
614 prop_binCRF_muon2016M_bin6
615 prop_binCRF_muon2018_bin7_TVX
616 prop_binCRF_electron2017_bin2_Fake
617 prop_binCRTT_electron2018_bin17
618 prop_binCRF_muon2018_bin5_TTTo2L2Nu
619 prop_binCROS_muon2018_bin17_Triboson
620 prop_binSR_electron2017_bin11_TVX
621 prop_binCROS_muon2016M_bin12
622 prop_binSR_muon2016M_bin9_Triboson
623 prop_binCROS_muon2017_bin8_TVX
624 prop_binCROS_muon2016M_bin15_TTTo2L2Nu
625 prop_binCRF_electron2018_bin16
626 prop_binCRF_electron2018_bin8
627 prop_binSR_electron2016M_bin10_WpWpJJ_QCD
628 prop_binCROS_muon2016M_bin15_WrongSign
629 prop_binSR_electron2017_bin14_ZZtoLep
630 prop_binCRF_electron2016M_bin15_sm
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



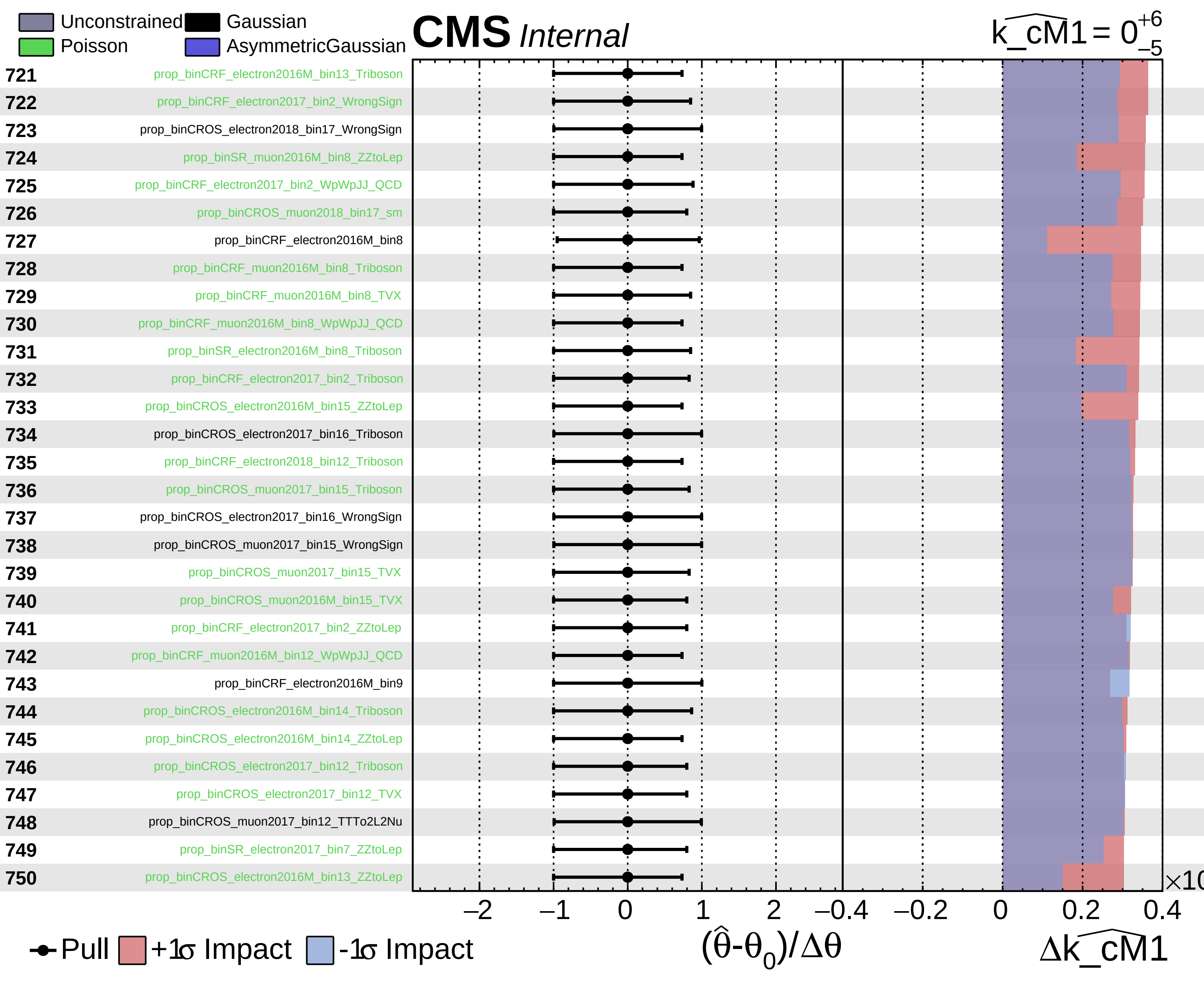
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
631 prop_binSR_electron2017_bin9_WrongSign
632 prop_binCRF_electron2018_bin12_Fake
633 prop_binSR_electron2018_bin11_Triboson
634 prop_binCRF_muon2016M_bin8_ZZtoLep
635 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
636 prop_binSR_electron2017_bin10_TVX
637 prop_binCROS_electron2018_bin14_sm
638 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
639 prop_binCRF_electron2017_bin11_TTTo2L2Nu
640 prop_binCROS_electron2016M_bin15_TVX
641 prop_binSR_muon2017_bin4_TVX
642 prop_binSR_electron2018_bin11_WpWpJJ_QCD
643 prop_binSR_electron2017_bin13_TVX
644 prop_binSR_muon2016M_bin13_TVX
645 prop_binCROS_electron2017_bin17_TVX
646 prop_binSR_electron2017_bin9_Triboson
647 prop_binSR_electron2017_bin7_TVX
648 prop_binSR_muon2018_bin6_WrongSign
649 prop_binSR_electron2017_bin7_WpWpJJ_QCD
650 prop_binSR_muon2016M_bin10_Triboson
651 prop_binSR_muon2017_bin4_WpWpJJ_QCD
652 prop_binCROS_muon2016M_bin17_TTTo2L2Nu
653 prop_binCROS_muon2018_bin16_Triboson
654 prop_binCROS_muon2018_bin5_WZ
655 prop_binCROS_electron2018_bin11_ZZtoLep
656 prop_binSR_electron2017_bin13_ZZtoLep
657 prop_binCRTT_muon2016M_bin2
658 prop_binCROS_electron2017_bin17_Triboson
659 prop_binCROS_muon2017_bin15_TTTo2L2Nu
660 prop_binCROS_electron2018_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

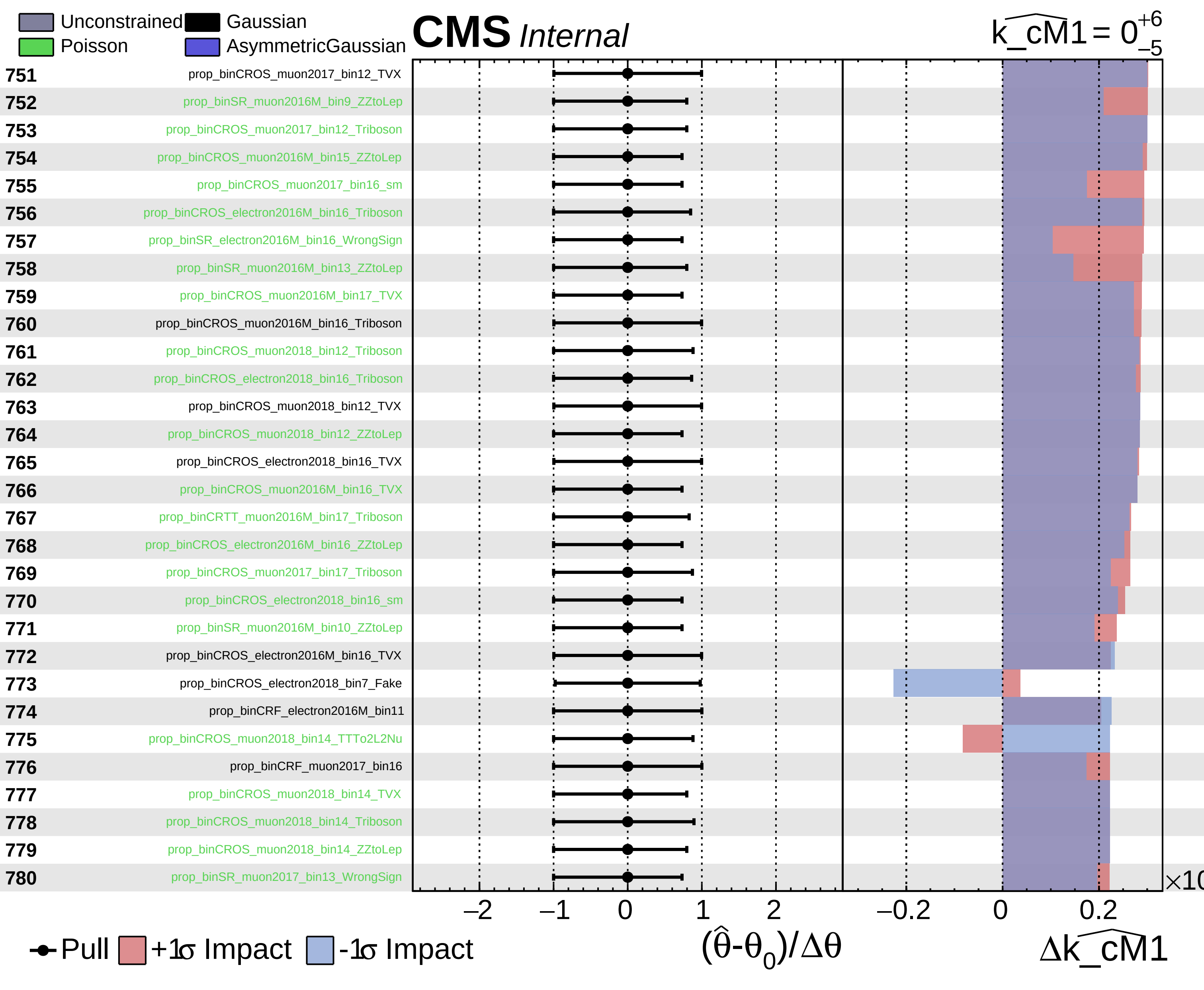


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
661 prop_binSR_electron2017_bin10_WrongSign
662 prop_binCROS_electron2016M_bin11_TVX
663 prop_binSR_muon2016M_bin10_TVX
664 prop_binSR_electron2017_bin10_Triboson
665 prop_binCRF_electron2018_bin11
666 prop_binSR_electron2017_bin7_Triboson
667 prop_binCROS_muon2016M_bin17_Triboson
668 prop_binSR_muon2016M_bin12_TVX
669 prop_binCRF_muon2018_bin5_TVX
670 prop_binSR_electron2017_bin10_WpWpJJ_QCD
671 prop_binSR_muon2018_bin12_ZZtoLep
672 prop_binCROS_electron2016M_bin11_ZZtoLep
673 prop_binSR_muon2017_bin4_Triboson
674 prop_binSR_electron2017_bin11_Triboson
675 prop_binCROS_electron2018_bin17_Triboson
676 prop_binCROS_muon2017_bin5_Triboson
677 prop_binCROS_electron2016M_bin12_ZZtoLep
678 prop_binSR_electron2016M_bin8_TVX
679 prop_binCRF_muon2016M_bin14
680 prop_binSR_muon2017_bin13_WpWpJJ_QCD
681 prop_binSR_electron2016M_bin11_TVX
682 prop_binCROS_muon2018_bin17_ZZtoLep
683 prop_binSR_muon2017_bin13_TVX
684 prop_binCROS_muon2018_bin5_sm
685 prop_binCROS_muon2017_bin8_ZZtoLep
686 prop_binCROS_muon2018_bin12_TTTo2L2Nu
687 prop_binCRF_electron2018_bin12_WrongSign
688 prop_binCRF_muon2018_bin5_Triboson
689 prop_binCROS_muon2017_bin5_sm
690 prop_binSR_electron2016M_bin12_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1

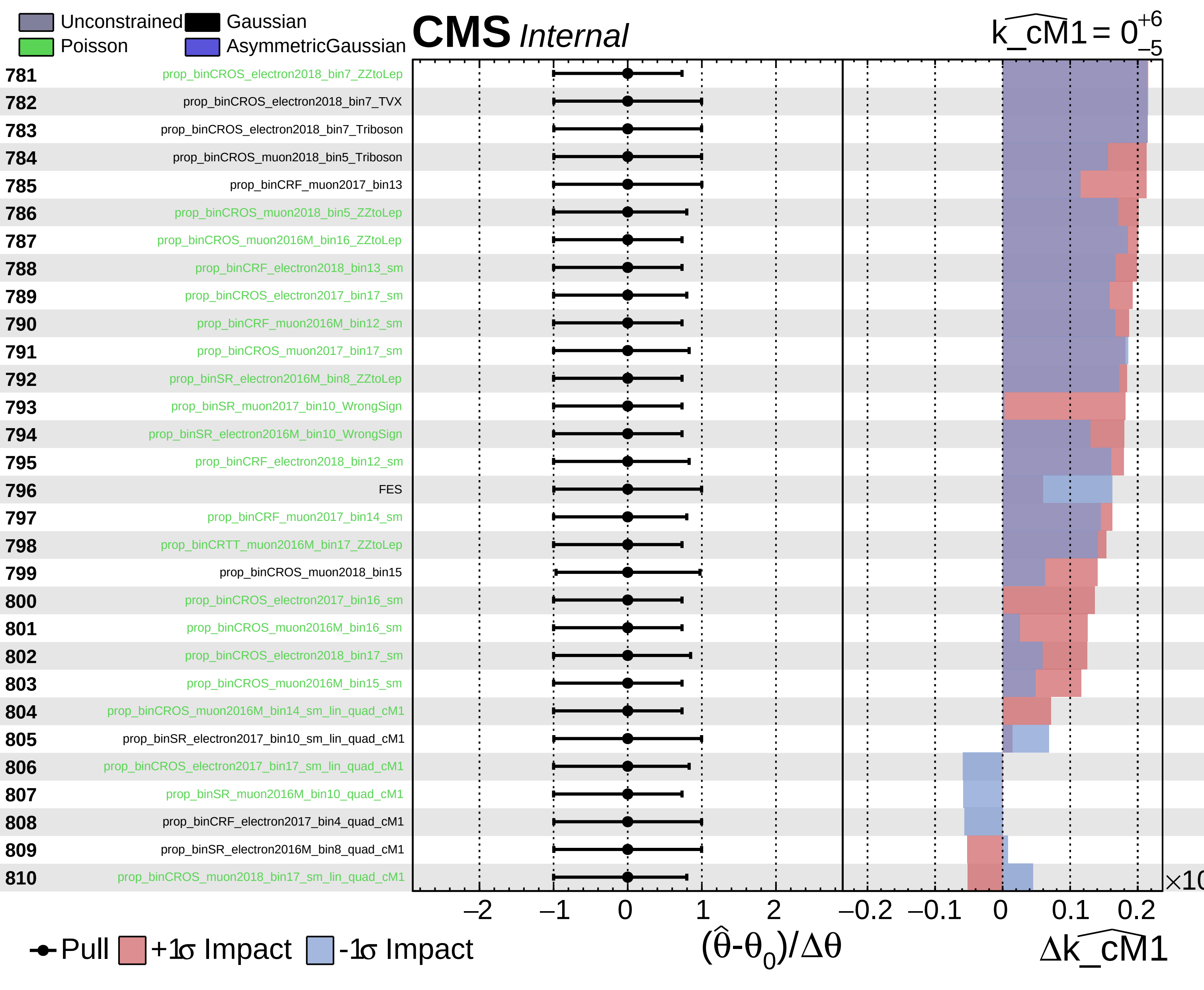


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
691 prop_binSR_muon2018_bin6_ZZtoLep
692 prop_binCROS_muon2018_bin17_TVX
693 prop_binCRF_muon2018_bin5_WpWpJJ_QCD
694 prop_binCRTT_muon2016M_bin17_TVX
695 prop_binSR_muon2017_bin13_Triboson
696 prop_binSR_electron2016M_bin8_WpWpJJ_QCD
697 prop_binCROS_electron2016M_bin17_WrongSign
698 prop_binSR_electron2016M_bin11_ZZtoLep
699 prop_binCRF_muon2017_bin8_ZZtoLep
700 prop_binCROS_muon2018_bin17_WrongSign
701 prop_binCRF_electron2018_bin14
702 prop_binSR_electron2016M_bin8_WrongSign
703 prop_binCROS_electron2016M_bin14_Fake
704 prop_binSR_electron2017_bin9_TVX
705 prop_binCROS_electron2017_bin17_WrongSign
706 prop_binCRF_muon2018_bin7_ZZtoLep
707 prop_binCRF_electron2018_bin12_TVX
708 prop_binCRF_electron2016M_bin13_TVX
709 prop_binCROS_muon2018_bin16_TVX
710 prop_binCROS_muon2016M_bin15_Triboson
711 prop_binCROS_electron2016M_bin17_sm
712 prop_binCRF_electron2017_bin2_TVX
713 prop_binCRF_muon2016M_bin12_ZZtoLep
714 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
715 prop_binCROS_muon2018_bin5_TVX
716 prop_binCRF_muon2018_bin7_WpWpJJ_QCD
717 prop_binCRF_electron2017_bin11_TVX
718 prop_binCROS_electron2017_bin16_TTTo2L2Nu
719 prop_binCRF_electron2018_bin13_WrongSign
720 prop_binCRF_electron2017_bin2_TTTo2L2Nu
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

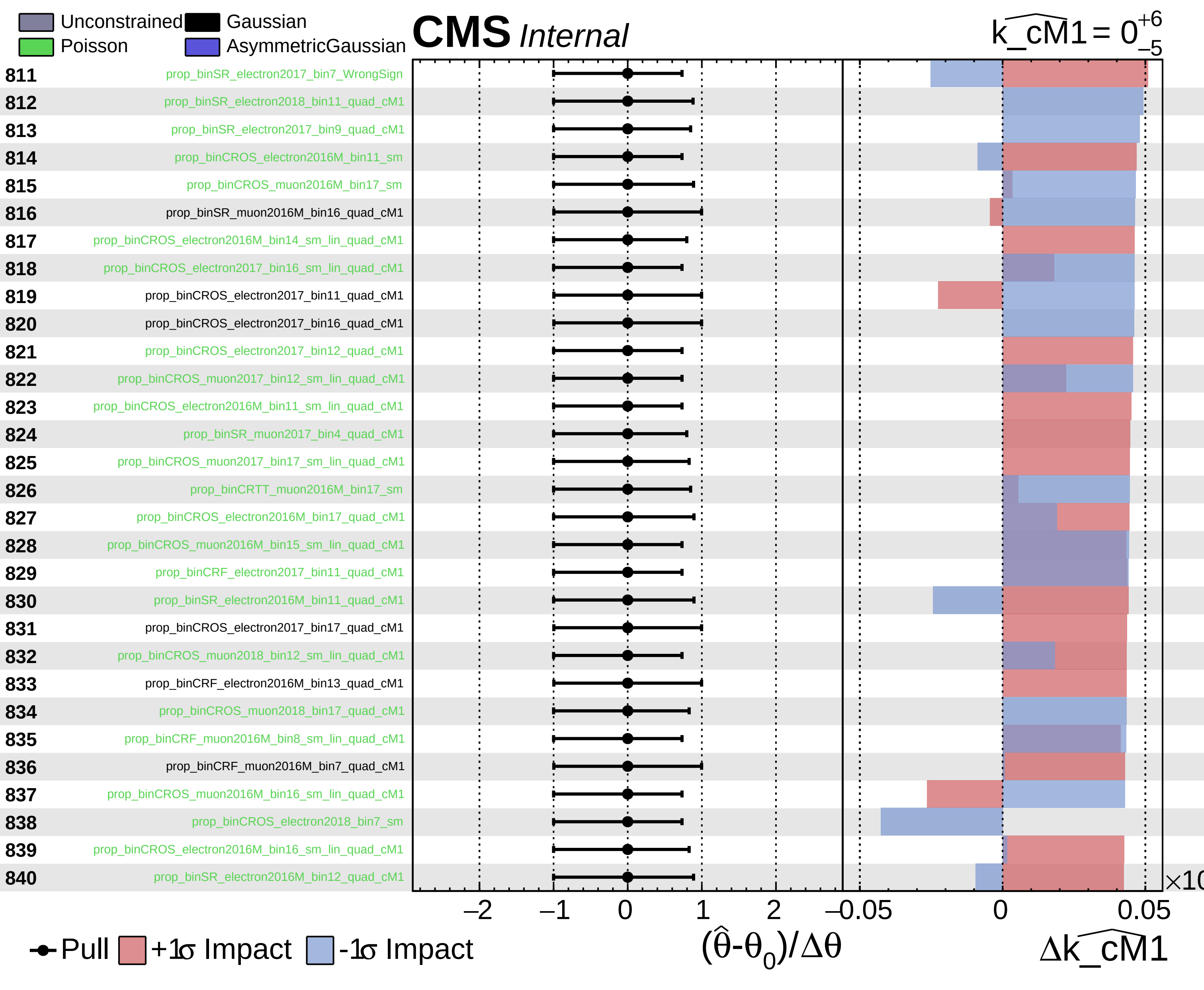

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
721 prop_binCRF_electron2016M_bin13_Triboson
722 prop_binCRF_electron2017_bin2_WrongSign
723 prop_binCROS_electron2018_bin17_WrongSign
724 prop_binSR_muon2016M_bin8_ZZtoLep
725 prop_binCRF_electron2017_bin2_WpWpJJ_QCD
726 prop_binCROS_muon2018_bin17_sm
727 prop_binCRF_electron2016M_bin8
728 prop_binCRF_muon2016M_bin8_Triboson
729 prop_binCRF_muon2016M_bin8_TVX
730 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
731 prop_binSR_electron2016M_bin8_Triboson
732 prop_binCRF_electron2017_bin2_Triboson
733 prop_binCROS_electron2016M_bin15_ZZtoLep
734 prop_binCROS_electron2017_bin16_Triboson
735 prop_binCRF_electron2018_bin12_Triboson
736 prop_binCROS_muon2017_bin15_Triboson
737 prop_binCROS_electron2017_bin16_WrongSign
738 prop_binCROS_muon2017_bin15_WrongSign
739 prop_binCROS_muon2017_bin15_TVX
740 prop_binCROS_muon2016M_bin15_TVX
741 prop_binCRF_electron2017_bin2_ZZtoLep
742 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
743 prop_binCRF_electron2016M_bin9
744 prop_binCROS_electron2016M_bin14_Triboson
745 prop_binCROS_electron2016M_bin14_ZZtoLep
746 prop_binCROS_electron2017_bin12_Triboson
747 prop_binCROS_electron2017_bin12_TVX
748 prop_binCROS_muon2017_bin12_TTTo2L2Nu
749 prop_binSR_electron2017_bin7_ZZtoLep
750 prop_binCROS_electron2016M_bin13_ZZtoLep
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1



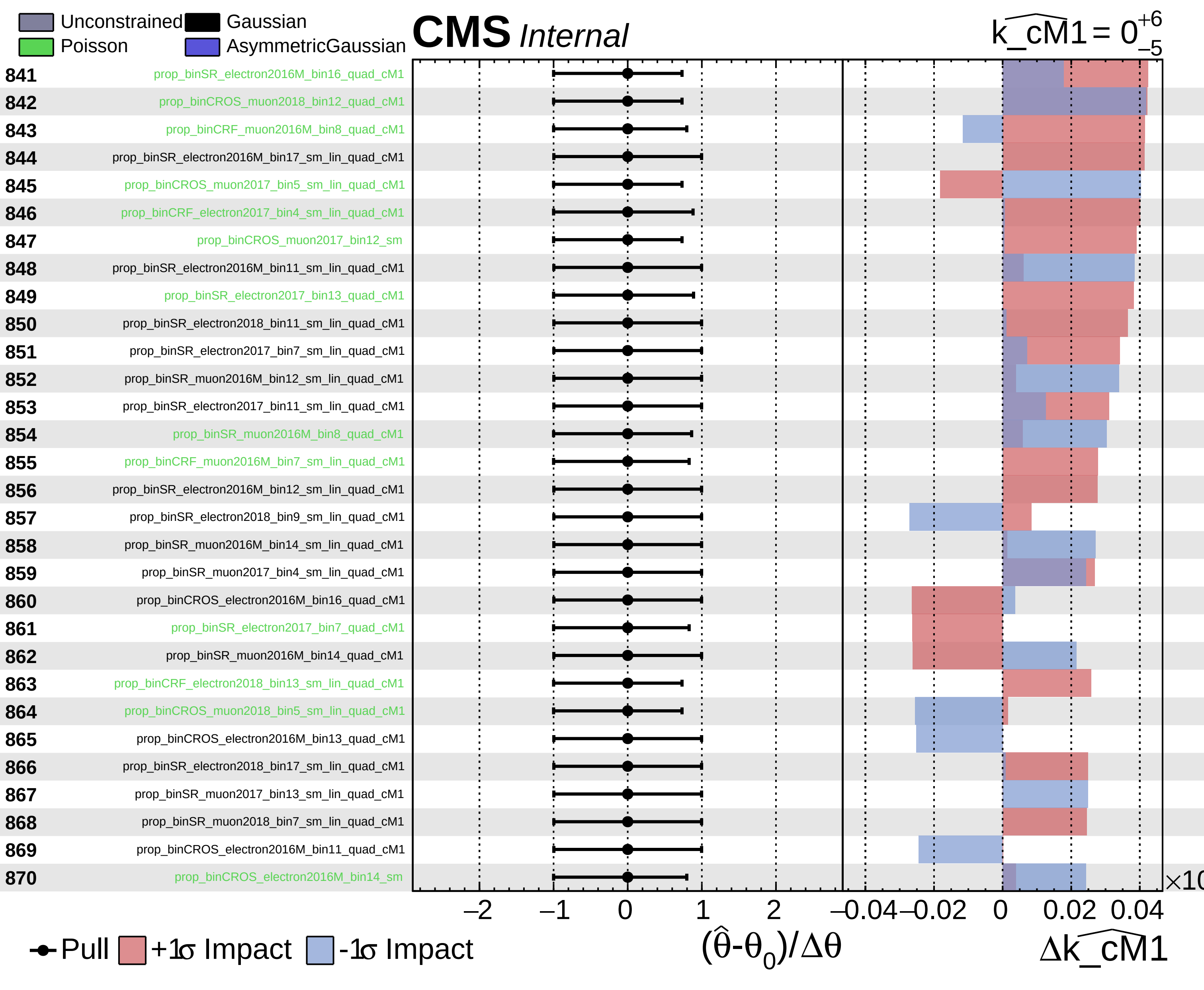
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
751 prop_binCROS_muon2017_bin12_TVX
752 prop_binSR_muon2016M_bin9_ZZtoLep
753 prop_binCROS_muon2017_bin12_Triboson
754 prop_binCROS_muon2016M_bin15_ZZtoLep
755 prop_binCROS_muon2017_bin16_sm
756 prop_binCROS_electron2016M_bin16_Triboson
757 prop_binSR_electron2016M_bin16_WrongSign
758 prop_binSR_muon2016M_bin13_ZZtoLep
759 prop_binCROS_muon2016M_bin17_TVX
760 prop_binCROS_muon2016M_bin16_Triboson
761 prop_binCROS_muon2018_bin12_Triboson
762 prop_binCROS_electron2018_bin16_Triboson
763 prop_binCROS_muon2018_bin12_TVX
764 prop_binCROS_muon2018_bin12_ZZtoLep
765 prop_binCROS_electron2018_bin16_TVX
766 prop_binCROS_muon2016M_bin16_TVX
767 prop_binCRTT_muon2016M_bin17_Triboson
768 prop_binCROS_electron2016M_bin16_ZZtoLep
769 prop_binCROS_muon2017_bin17_Triboson
770 prop_binCROS_electron2018_bin16_sm
771 prop_binSR_muon2016M_bin10_ZZtoLep
772 prop_binCROS_electron2016M_bin16_TVX
773 prop_binCROS_electron2018_bin7_Fake
774 prop_binCRF_electron2016M_bin11
775 prop_binCROS_muon2018_bin14_TTTo2L2Nu
776 prop_binCRF_muon2017_bin16
777 prop_binCROS_muon2018_bin14_TVX
778 prop_binCROS_muon2018_bin14_Triboson
779 prop_binCROS_muon2018_bin14_ZZtoLep
780 prop_binSR_muon2017_bin13_WrongSign
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

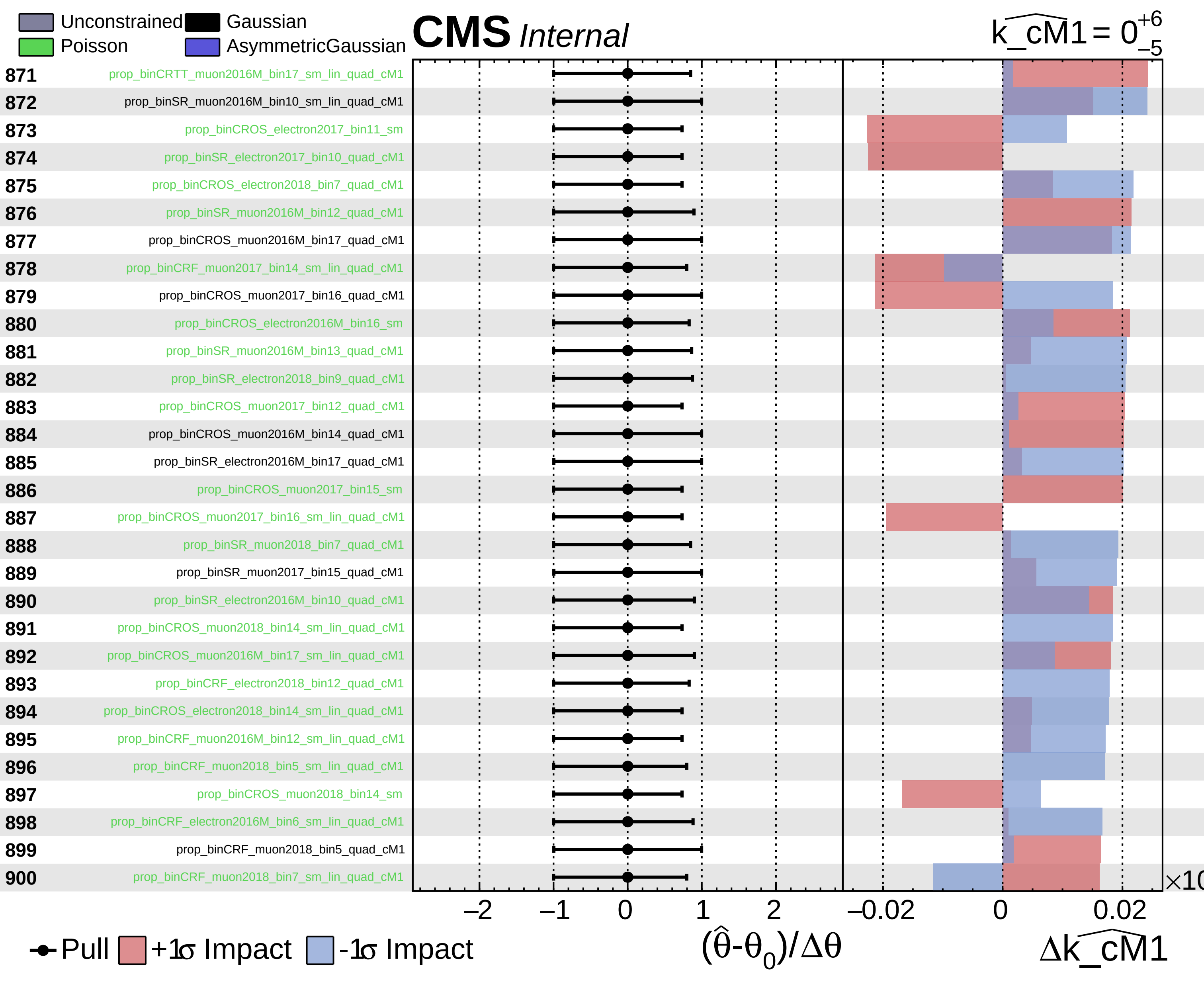


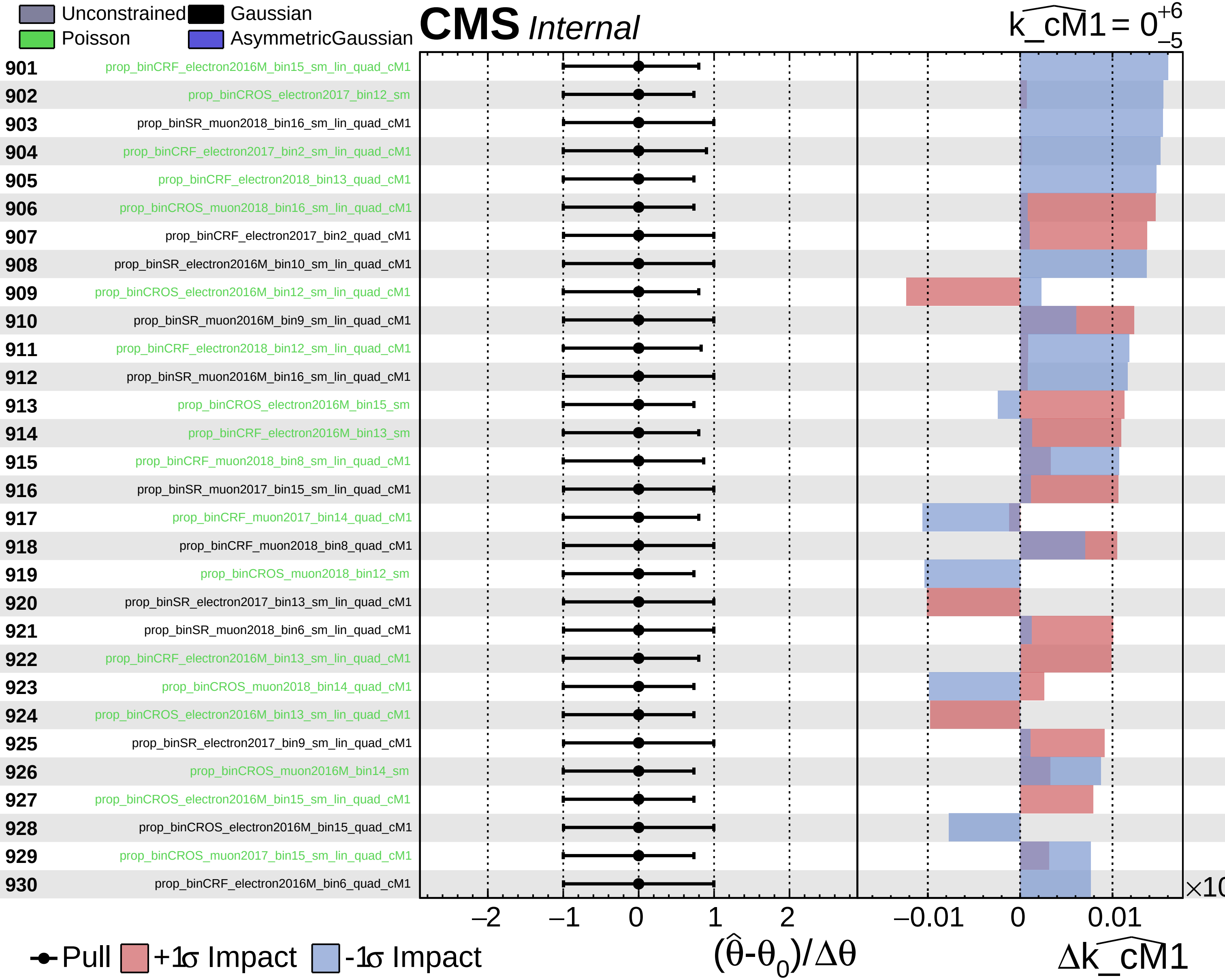
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
781 prop_binCROS_electron2018_bin7_ZZtoLep
782 prop_binCROS_electron2018_bin7_TVX
783 prop_binCROS_electron2018_bin7_Triboson
784 prop_binCROS_muon2018_bin5_Triboson
785 prop_binCRF_muon2017_bin13
786 prop_binCROS_muon2018_bin5_ZZtoLep
787 prop_binCROS_muon2016M_bin16_ZZtoLep
788 prop_binCRF_electron2018_bin13_sm
789 prop_binCROS_electron2017_bin17_sm
790 prop_binCRF_muon2016M_bin12_sm
791 prop_binCROS_muon2017_bin17_sm
792 prop_binSR_electron2016M_bin8_ZZtoLep
793 prop_binSR_muon2017_bin10_WrongSign
794 prop_binSR_electron2016M_bin10_WrongSign
795 prop_binCRF_electron2018_bin12_sm
796 FES
797 prop_binCRF_muon2017_bin14_sm
798 prop_binCRTT_muon2016M_bin17_ZZtoLep
799 prop_binCROS_muon2018_bin15
800 prop_binCROS_electron2017_bin16_sm
801 prop_binCROS_muon2016M_bin16_sm
802 prop_binCROS_electron2018_bin17_sm
803 prop_binCROS_muon2016M_bin15_sm
804 prop_binCROS_muon2016M_bin14_sm_lin_quad_cM1
805 prop_binSR_electron2017_bin10_sm_lin_quad_cM1
806 prop_binCROS_electron2017_bin17_sm_lin_quad_cM1
807 prop_binSR_muon2016M_bin10_quad_cM1
808 prop_binCRF_electron2017_bin4_quad_cM1
809 prop_binSR_electron2016M_bin8_quad_cM1
810 prop_binCROS_muon2018_bin17_sm_lin_quad_cM1
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



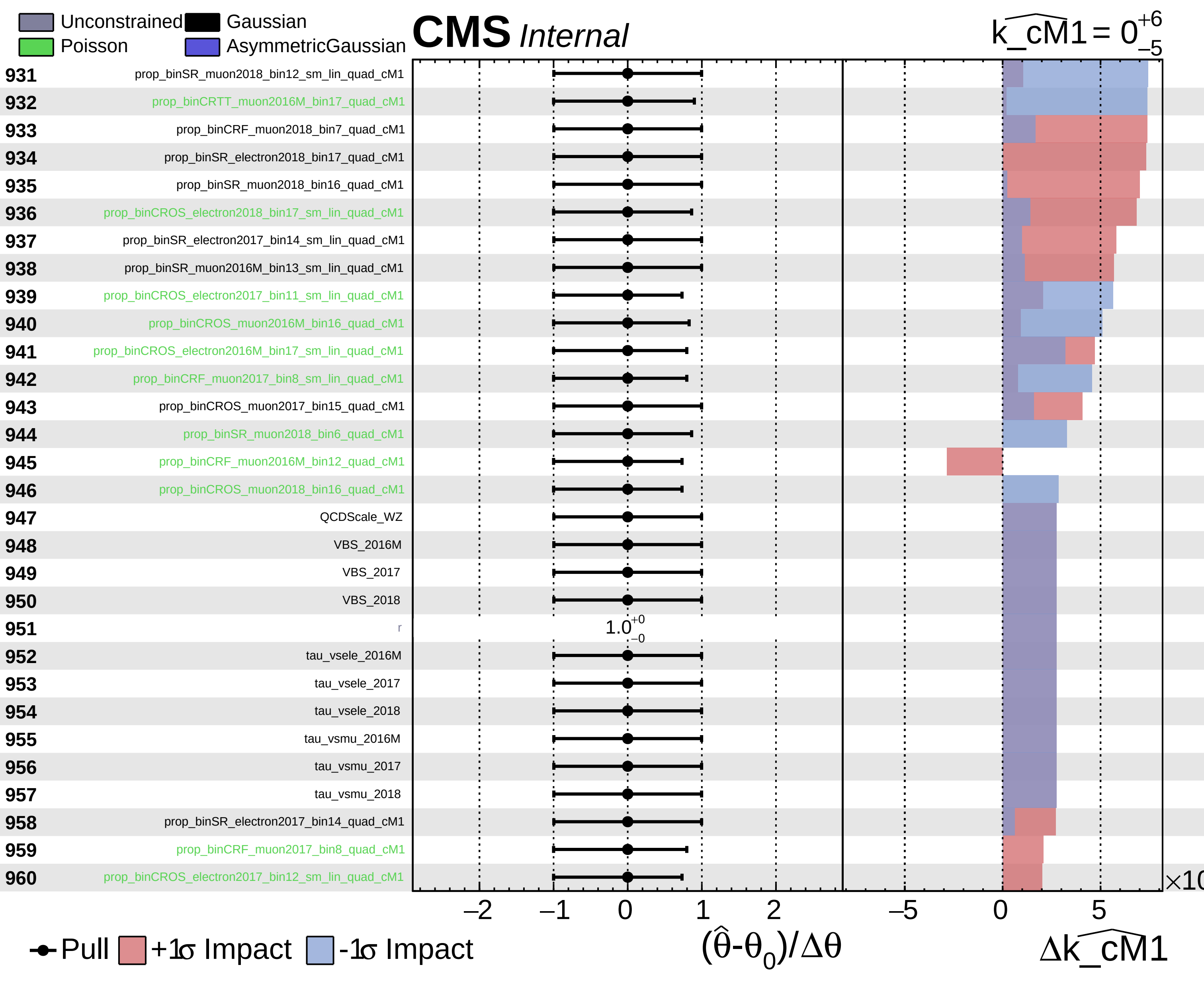
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
811 prop_binSR_electron2017_bin7_WrongSign
812 prop_binSR_electron2018_bin11_quad_cM1
813 prop_binSR_electron2017_bin9_quad_cM1
814 prop_binCROS_electron2016M_bin11_sm
815 prop_binCROS_muon2016M_bin17_sm
816 prop_binSR_muon2016M_bin16_quad_cM1
817 prop_binCROS_electron2016M_bin14_sm_lin_quad_cM1
818 prop_binCROS_electron2017_bin16_sm_lin_quad_cM1
819 prop_binCROS_electron2017_bin11_quad_cM1
820 prop_binCROS_electron2017_bin16_quad_cM1
821 prop_binCROS_electron2017_bin12_quad_cM1
822 prop_binCROS_muon2017_bin12_sm_lin_quad_cM1
823 prop_binCROS_electron2016M_bin11_sm_lin_quad_cM1
824 prop_binSR_muon2017_bin4_quad_cM1
825 prop_binCROS_muon2017_bin17_sm_lin_quad_cM1
826 prop_binCRTT_muon2016M_bin17_sm
827 prop_binCROS_electron2016M_bin17_quad_cM1
828 prop_binCROS_muon2016M_bin15_sm_lin_quad_cM1
829 prop_binCRF_electron2017_bin11_quad_cM1
830 prop_binSR_electron2016M_bin11_quad_cM1
831 prop_binCROS_electron2017_bin17_quad_cM1
832 prop_binCROS_muon2018_bin12_sm_lin_quad_cM1
833 prop_binCRF_electron2016M_bin13_quad_cM1
834 prop_binCROS_muon2018_bin17_quad_cM1
835 prop_binCRF_muon2016M_bin8_sm_lin_quad_cM1
836 prop_binCRF_muon2016M_bin7_quad_cM1
837 prop_binCROS_muon2016M_bin16_sm_lin_quad_cM1
838 prop_binCROS_electron2018_bin7_sm
839 prop_binCROS_electron2016M_bin16_sm_lin_quad_cM1
840 prop_binSR_electron2016M_bin12_quad_cM1
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

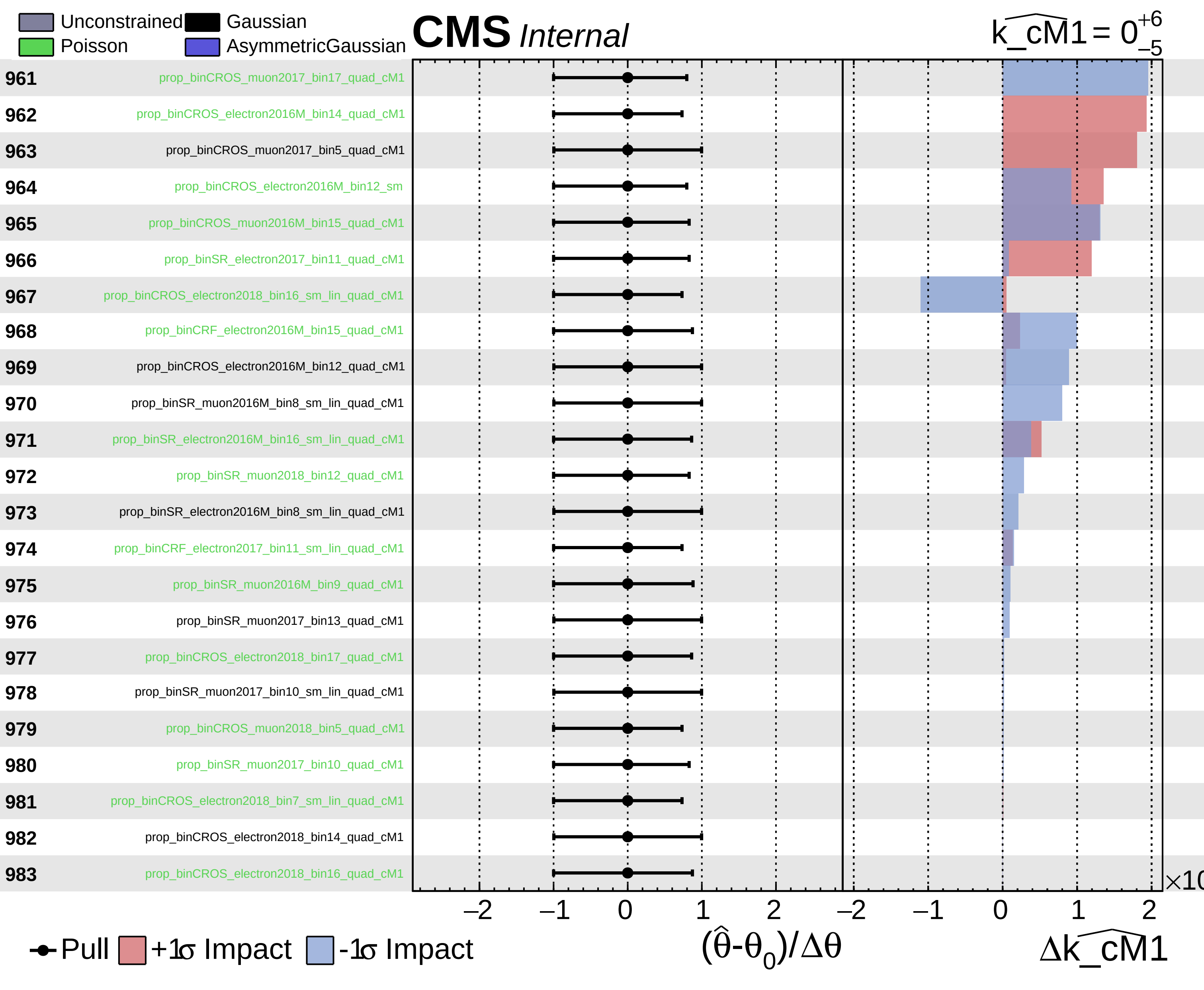

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
841 prop_binSR_electron2016M_bin16_quad_cM1
842 prop_binCROS_muon2018_bin12_quad_cM1
843 prop_binCRF_muon2016M_bin8_quad_cM1
844 prop_binSR_electron2016M_bin17_sm_lin_quad_cM1
845 prop_binCROS_muon2017_bin5_sm_lin_quad_cM1
846 prop_binCRF_electron2017_bin4_sm_lin_quad_cM1
847 prop_binCROS_muon2017_bin12_sm
848 prop_binSR_electron2016M_bin11_sm_lin_quad_cM1
849 prop_binSR_electron2017_bin13_quad_cM1
850 prop_binSR_electron2018_bin11_sm_lin_quad_cM1
851 prop_binSR_electron2017_bin7_sm_lin_quad_cM1
852 prop_binSR_muon2016M_bin12_sm_lin_quad_cM1
853 prop_binSR_electron2017_bin11_sm_lin_quad_cM1
854 prop_binSR_muon2016M_bin8_quad_cM1
855 prop_binCRF_muon2016M_bin7_sm_lin_quad_cM1
856 prop_binSR_electron2016M_bin12_sm_lin_quad_cM1
857 prop_binSR_electron2018_bin9_sm_lin_quad_cM1
858 prop_binSR_muon2016M_bin14_sm_lin_quad_cM1
859 prop_binSR_muon2017_bin4_sm_lin_quad_cM1
860 prop_binCROS_electron2016M_bin16_quad_cM1
861 prop_binSR_electron2017_bin7_quad_cM1
862 prop_binSR_muon2016M_bin14_quad_cM1
863 prop_binCRF_electron2018_bin13_sm_lin_quad_cM1
864 prop_binCROS_muon2018_bin5_sm_lin_quad_cM1
865 prop_binCROS_electron2016M_bin13_quad_cM1
866 prop_binSR_electron2018_bin17_sm_lin_quad_cM1
867 prop_binSR_muon2017_bin13_sm_lin_quad_cM1
868 prop_binSR_muon2018_bin7_sm_lin_quad_cM1
869 prop_binCROS_electron2016M_bin11_quad_cM1
870 prop_binCROS_electron2016M_bin14_sm
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
871 prop_binCRTT_muon2016M_bin17_sm_lin_quad_cM1
872 prop_binSR_muon2016M_bin10_sm_lin_quad_cM1
873 prop_binCROS_electron2017_bin11_sm
874 prop_binSR_electron2017_bin10_quad_cM1
875 prop_binCROS_electron2018_bin7_quad_cM1
876 prop_binSR_muon2016M_bin12_quad_cM1
877 prop_binCROS_muon2016M_bin17_quad_cM1
878 prop_binCRF_muon2017_bin14_sm_lin_quad_cM1
879 prop_binCROS_muon2017_bin16_quad_cM1
880 prop_binCROS_electron2016M_bin16_sm
881 prop_binSR_muon2016M_bin13_quad_cM1
882 prop_binSR_electron2018_bin9_quad_cM1
883 prop_binCROS_muon2017_bin12_quad_cM1
884 prop_binCROS_muon2016M_bin14_quad_cM1
885 prop_binSR_electron2016M_bin17_quad_cM1
886 prop_binCROS_muon2017_bin15_sm
887 prop_binCROS_muon2017_bin16_sm_lin_quad_cM1
888 prop_binSR_muon2018_bin7_quad_cM1
889 prop_binSR_muon2017_bin15_quad_cM1
890 prop_binSR_electron2016M_bin10_quad_cM1
891 prop_binCROS_muon2018_bin14_sm_lin_quad_cM1
892 prop_binCROS_muon2016M_bin17_sm_lin_quad_cM1
893 prop_binCRF_electron2018_bin12_quad_cM1
894 prop_binCROS_electron2018_bin14_sm_lin_quad_cM1
895 prop_binCRF_muon2016M_bin12_sm_lin_quad_cM1
896 prop_binCRF_muon2018_bin5_sm_lin_quad_cM1
897 prop_binCROS_muon2018_bin14_sm
898 prop_binCRF_electron2016M_bin6_sm_lin_quad_cM1
899 prop_binCRF_muon2018_bin5_quad_cM1
900 prop_binCRF_muon2018_bin7_sm_lin_quad_cM1
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
901 prop_binCRF_electron2016M_bin15_sm_lin_quad_cM1
902 prop_binCROS_electron2017_bin12_sm
903 prop_binSR_muon2018_bin16_sm_lin_quad_cM1
904 prop_binCRF_electron2017_bin2_sm_lin_quad_cM1
905 prop_binCRF_electron2018_bin13_quad_cM1
906 prop_binCROS_muon2018_bin16_sm_lin_quad_cM1
907 prop_binCRF_electron2017_bin2_quad_cM1
908 prop_binSR_electron2016M_bin10_sm_lin_quad_cM1
909 prop_binCROS_electron2016M_bin12_sm_lin_quad_cM1
910 prop_binSR_muon2016M_bin9_sm_lin_quad_cM1
911 prop_binCRF_electron2018_bin12_sm_lin_quad_cM1
912 prop_binSR_muon2016M_bin16_sm_lin_quad_cM1
913 prop_binCROS_electron2016M_bin15_sm
914 prop_binCRF_electron2016M_bin13_sm
915 prop_binCRF_muon2018_bin8_sm_lin_quad_cM1
916 prop_binSR_muon2017_bin15_sm_lin_quad_cM1
917 prop_binCRF_muon2017_bin14_quad_cM1
918 prop_binCRF_muon2018_bin8_quad_cM1
919 prop_binCROS_muon2018_bin12_sm
920 prop_binSR_electron2017_bin13_sm_lin_quad_cM1
921 prop_binSR_muon2018_bin6_sm_lin_quad_cM1
922 prop_binCRF_electron2016M_bin13_sm_lin_quad_cM1
923 prop_binCROS_muon2018_bin14_quad_cM1
924 prop_binCROS_electron2016M_bin13_sm_lin_quad_cM1
925 prop_binSR_electron2017_bin9_sm_lin_quad_cM1
926 prop_binCROS_muon2016M_bin14_sm
927 prop_binCROS_electron2016M_bin15_sm_lin_quad_cM1
928 prop_binCROS_electron2016M_bin15_quad_cM1
929 prop_binCROS_muon2017_bin15_sm_lin_quad_cM1
930 prop_binCRF_electron2016M_bin6_quad_cM1
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 0 +6 −5
931 prop_binSR_muon2018_bin12_sm_lin_quad_cM1
932 prop_binCRTT_muon2016M_bin17_quad_cM1
933 prop_binCRF_muon2018_bin7_quad_cM1
934 prop_binSR_electron2018_bin17_quad_cM1
935 prop_binSR_muon2018_bin16_quad_cM1
936 prop_binCROS_electron2018_bin17_sm_lin_quad_cM1
937 prop_binSR_electron2017_bin14_sm_lin_quad_cM1
938 prop_binSR_muon2016M_bin13_sm_lin_quad_cM1
939 prop_binCROS_electron2017_bin11_sm_lin_quad_cM1
940 prop_binCROS_muon2016M_bin16_quad_cM1
941 prop_binCROS_electron2016M_bin17_sm_lin_quad_cM1
942 prop_binCRF_muon2017_bin8_sm_lin_quad_cM1
943 prop_binCROS_muon2017_bin15_quad_cM1
944 prop_binSR_muon2018_bin6_quad_cM1
945 prop_binCRF_muon2016M_bin12_quad_cM1
946 prop_binCROS_muon2018_bin16_quad_cM1
947 QCDScale_WZ
948 VBS_2016M
949 VBS_2017
950 VBS_2018
951 r
1.0 +0 −0
952 tau_vsele_2016M
953 tau_vsele_2017
954 tau_vsele_2018
955 tau_vsmu_2016M
956 tau_vsmu_2017
957 tau_vsmu_2018
958 prop_binSR_electron2017_bin14_quad_cM1
959 prop_binCRF_muon2017_bin8_quad_cM1
960 prop_binCROS_electron2017_bin12_sm_lin_quad_cM1
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 0^{+6}_{-5}$
961 prop_binCROS_muon2017_bin17_quad_cM1
962 prop_binCROS_electron2016M_bin14_quad_cM1
963 prop_binCROS_muon2017_bin5_quad_cM1
964 prop_binCROS_electron2016M_bin12_sm
965 prop_binCROS_muon2016M_bin15_quad_cM1
966 prop_binSR_electron2017_bin11_quad_cM1
967 prop_binCROS_electron2018_bin16_sm_lin_quad_cM1
968 prop_binCRF_electron2016M_bin15_quad_cM1
969 prop_binCROS_electron2016M_bin12_quad_cM1
970 prop_binSR_muon2016M_bin8_sm_lin_quad_cM1
971 prop_binSR_electron2016M_bin16_sm_lin_quad_cM1
972 prop_binSR_muon2018_bin12_quad_cM1
973 prop_binSR_electron2016M_bin8_sm_lin_quad_cM1
974 prop_binCRF_electron2017_bin11_sm_lin_quad_cM1
975 prop_binSR_muon2016M_bin9_quad_cM1
976 prop_binSR_muon2017_bin13_quad_cM1
977 prop_binCROS_electron2018_bin17_quad_cM1
978 prop_binSR_muon2017_bin10_sm_lin_quad_cM1
979 prop_binCROS_muon2018_bin5_quad_cM1
980 prop_binSR_muon2017_bin10_quad_cM1
981 prop_binCROS_electron2018_bin7_sm_lin_quad_cM1
982 prop_binCROS_electron2018_bin14_quad_cM1
983 prop_binCROS_electron2018_bin16_quad_cM1
−2 −1 0 1 2
−2 −1 0 1 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$